南京大学
NANJING UNIVERSITY

计算机视觉表征与识别
Chapter 6: Edges and Boundaries

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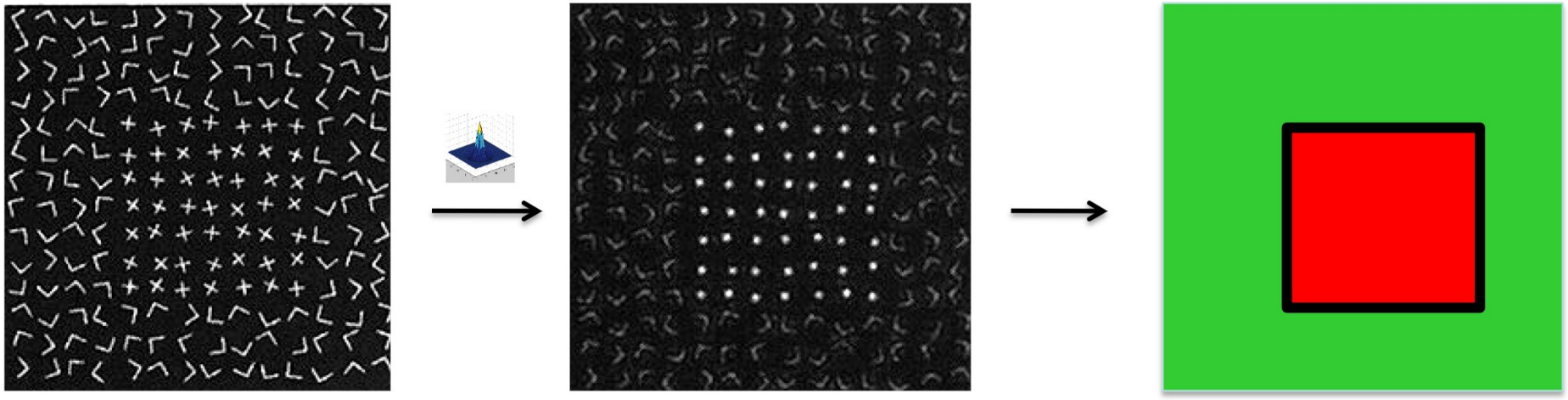
媒体计算课题组

http://mcg.nju.edu.cn/

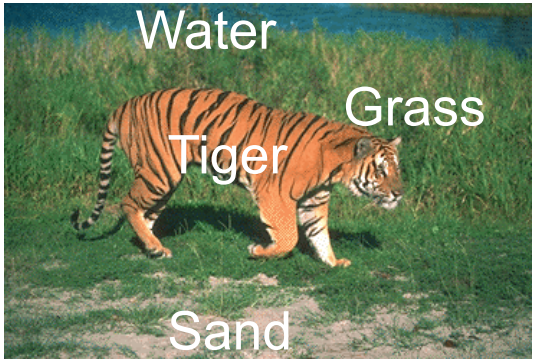
From texture to edge/segmentation

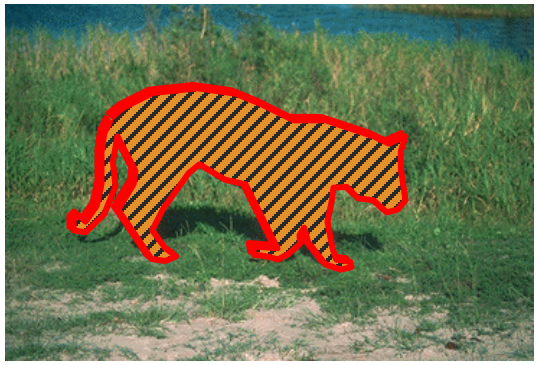


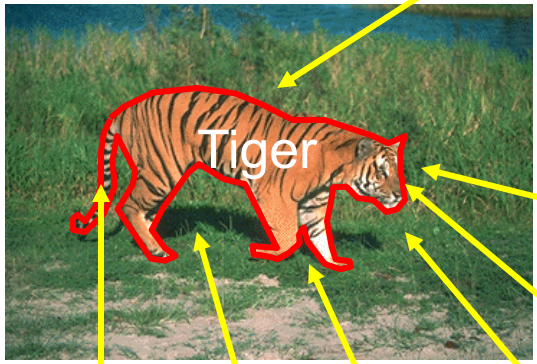
Low level Vision

Middle level Vision

From Pixels to Perception







back
head
eye
mouth
legs
tail
shadow

Mid-level operations of Segmentation and Grouping

Figure / Ground

Finding groups of pixels that go together (parts, objects, textures, holes)

Predicted scene categories[?]:

forest - broadleaf (0.498), swimming hole (0.402), bayou (0.062)

Predicted scene categories[?]:

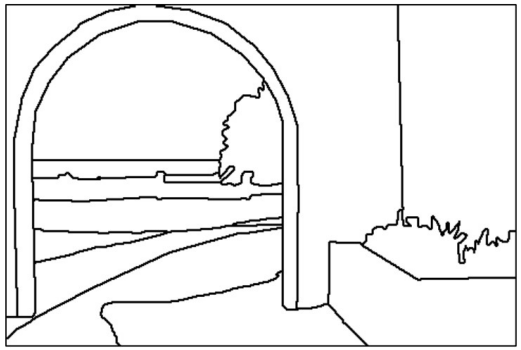
forest - broadleaf (0.979)

Figure / Ground

Finding groups of pixels that go together (parts, objects, textures, holes)

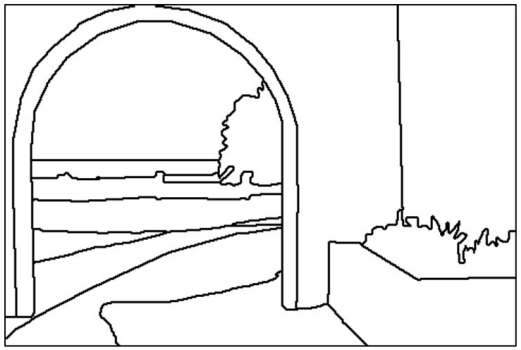
http://twistedsifter.com/2015/03/mind-bending-optical-illusion-paintings-by-rob-gonsalves/

Regions

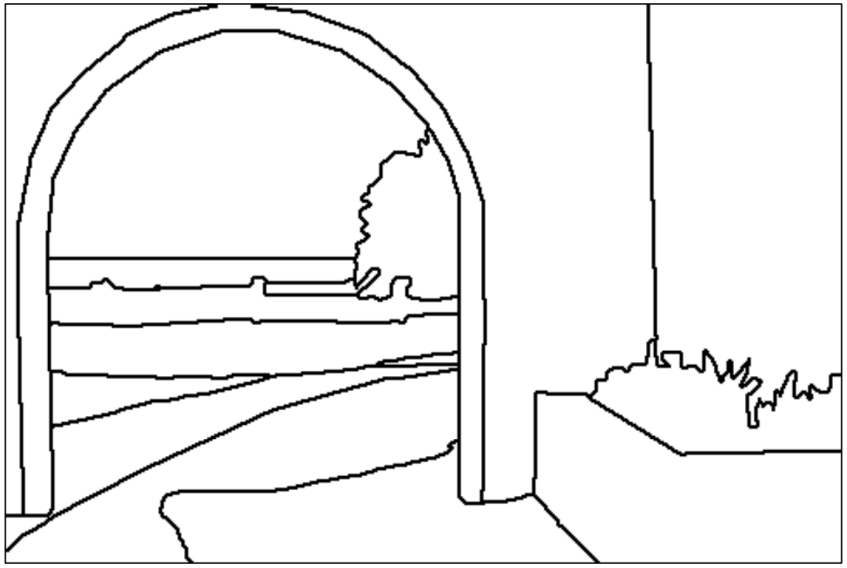


Decomposing image into K connected regions
(Clustering task)

Edges/Contour



HxWx {0,1} classification problem

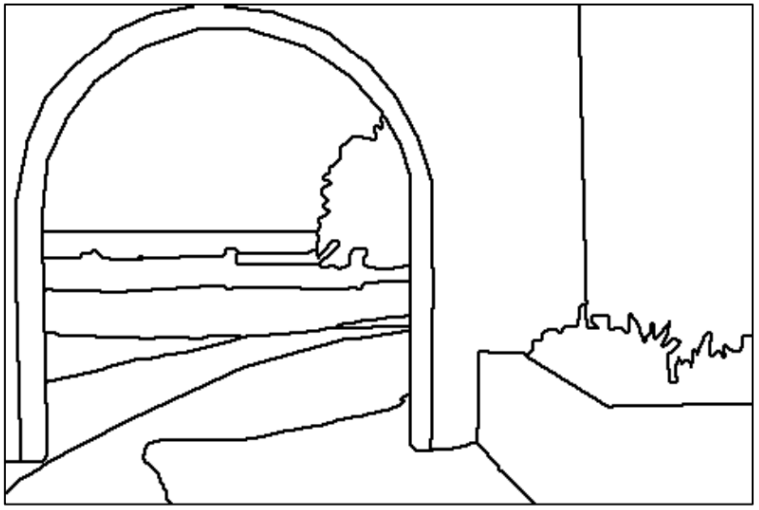


Are the tasks equivalent

Segmentation

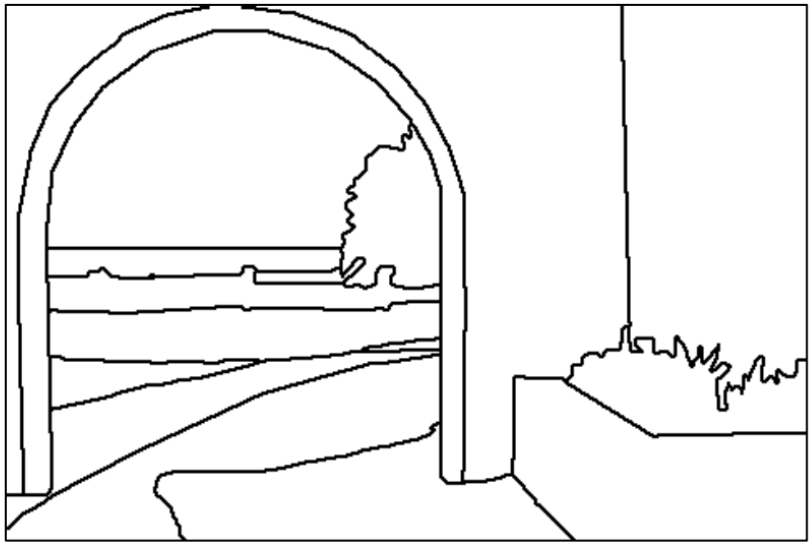
?

Boundaries



Are the tasks equivalent

Boundaries



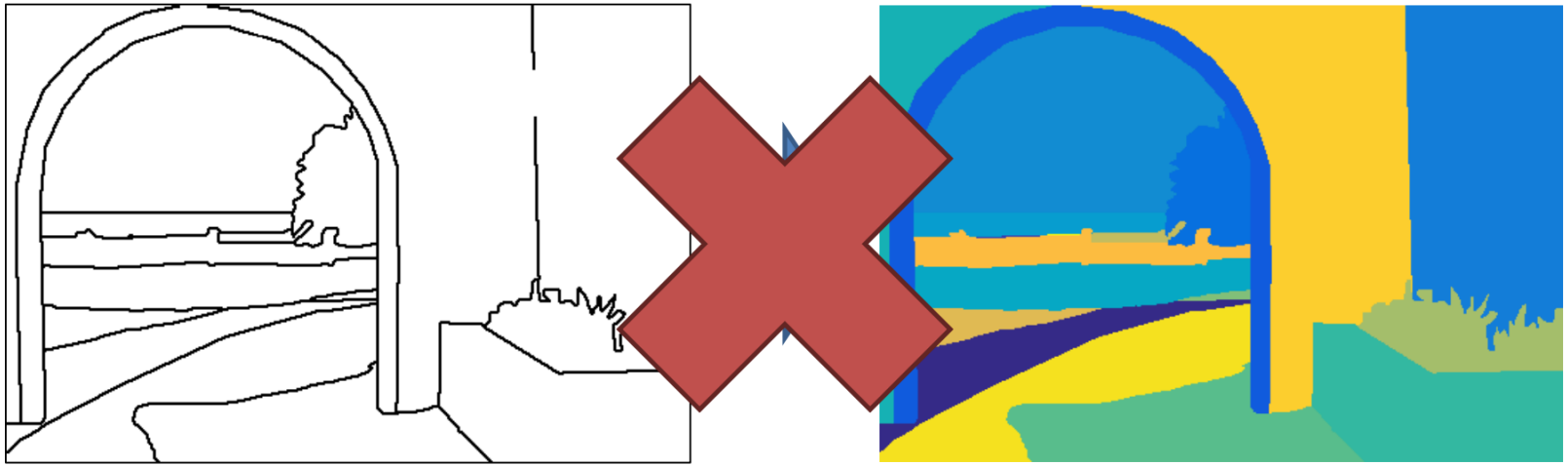
?

Segmentation

Are the tasks equivalent

Boundaries

Segmentation



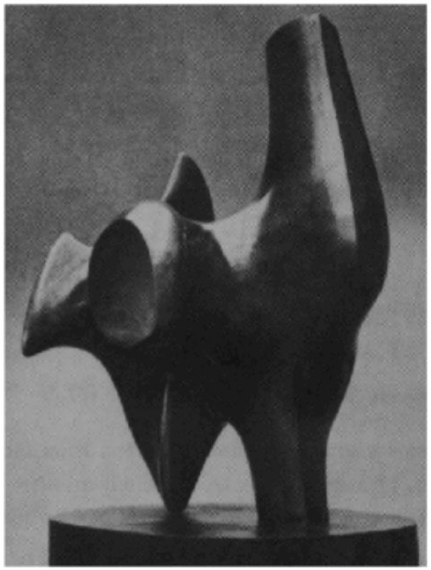
Contours have to be closed!

Today's class

- **Introduction to edge detection**
- Gradients and edges
- Canny edge detector
- Object contour
- Pb edge detector
- Straight line detection

First problem: edge detection

- Goal: compute something like a line drawing of a scene



Edge detection

- **Goal**: map image from 2d array of pixels to a set of curves or line segments or contours.

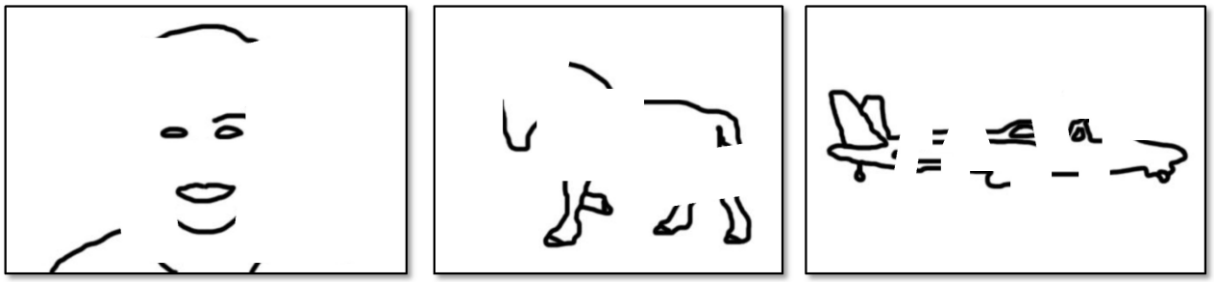


Figure from J. Shotton et al., PAMI 2007

- **Main idea**: look for strong gradients, post-process

Why

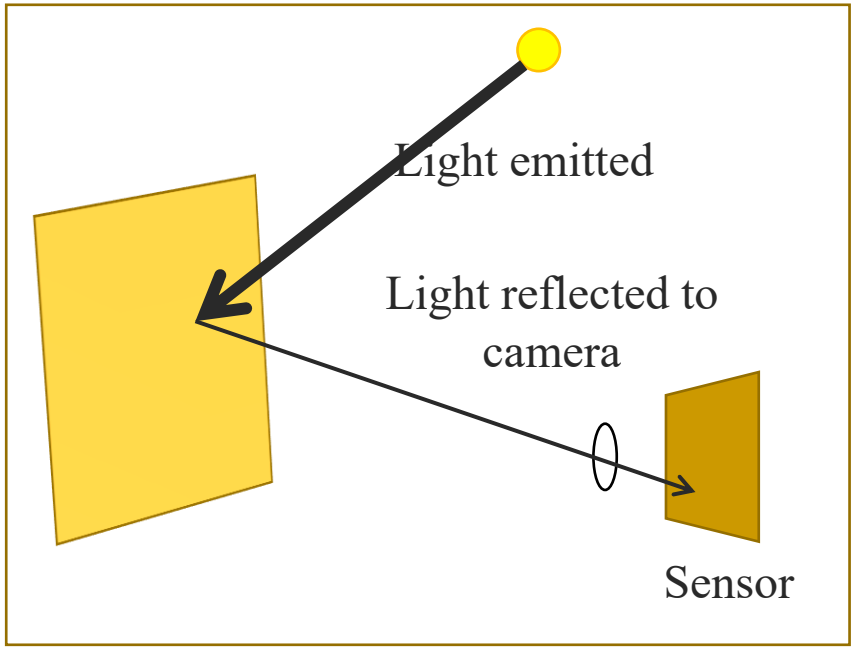
- Edges reflect intrinsic properties of a scene
  - Capture shape information
  - Independent of illumination
- Our human visual system does something like this
- Good initial step for solving other problems
  - Recognition, tracking, etc.

Issues

- No precise problem formulation
- Much harder than it seems to be
- Edge detectors usually work by detecting “big changes” in image intensity
- Boundary is contour in the image plane that represents a change in pixel ownership from object or surface to another.

Recall: How much light is recorded

- Major factors
  - Illumination strength and direction
  - Surface geometry
  - Surface material
  - Nearby surfaces
  - Camera gain/exposure



What causes an edge?

Reflectance change: appearance information, texture

Depth discontinuity: object boundary

Cast shadows

Change in surface orientation: shape

Slide credit: Kristen Grauman

Today's class

- Introduction to edge detection
- **Gradients and edges**
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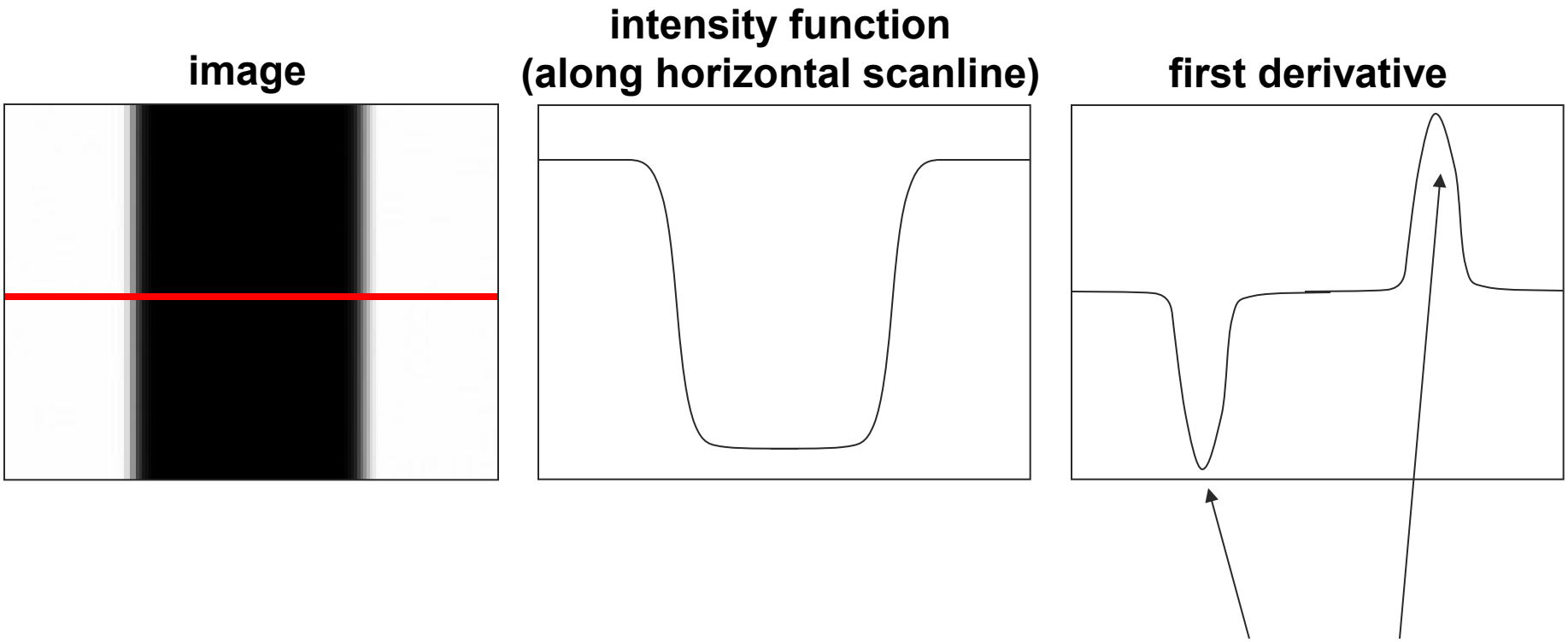
Derivatives and edges

An edge is a place of rapid change in the image intensity function.



Source: L. Lazebnik

Derivatives with convolution

For 2D function, f(x,y), the partial derivative is:

$$\frac{\partial f(x,y)}{\partial x} = \lim_{\varepsilon \to 0} \frac{f(x+\varepsilon, y) - f(x,y)}{\varepsilon}$$

For discrete data, we can approximate using finite differences:

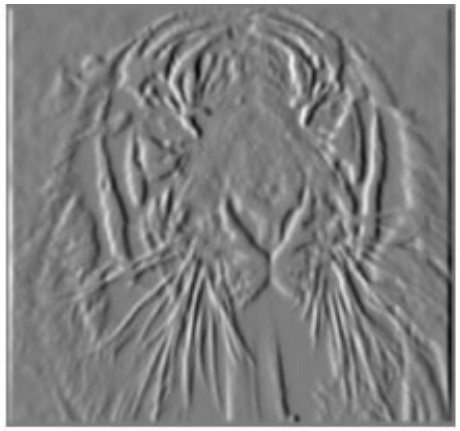
$$\frac{\partial f(x,y)}{\partial x} \approx \frac{f(x+1, y) - f(x,y)}{1}$$

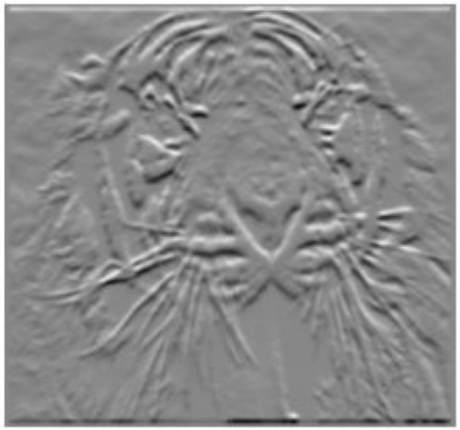
To implement above as convolution, what would be the associated filter?

Slide credit: Kristen Grauman

Partial derivatives of an image

$$\frac{\partial f(x,y)}{\partial x}$$





$$\frac{\partial f(x,y)}{\partial y}$$

-1	1

-1
1

? **or**

1
-1

Which shows changes with respect to x?

(showing filters for correlation)

Finite difference filters

- Other approximations of derivative filters exist:

Prewitt: $M_x =$

-1	0	1
-1	0	1
-1	0	1

; $M_y =$

1	1	1
0	0	0
-1	-1	-1

Sobel: $M_x =$

-1	0	1
-2	0	2
-1	0	1

; $M_y =$

1	2	1
0	0	0
-1	-2	-1

Roberts: $M_x =$

0	1
-1	0

; $M_y =$

1	0
0	-1

Source: K. Grauman

Image gradient

The gradient of an image:

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right]$$

The gradient points in the direction of most rapid change in intensity

$$\nabla f = \left[\frac{\partial f}{\partial x}, 0\right]$$

$$\nabla f = \left[0, \frac{\partial f}{\partial y}\right]$$

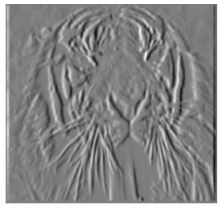
$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right]$$

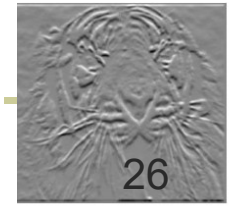
The **gradient direction** (orientation of edge normal) is given by:

$$\theta = \tan^{-1}\left(\frac{\partial f}{\partial y} / \frac{\partial f}{\partial x}\right)$$

The **edge strength** is given by the gradient magnitude

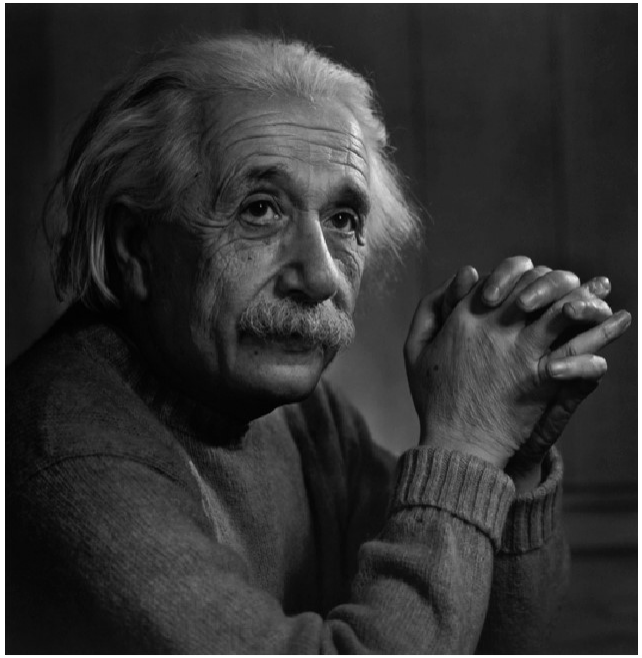
$$\|\nabla f\| = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}$$

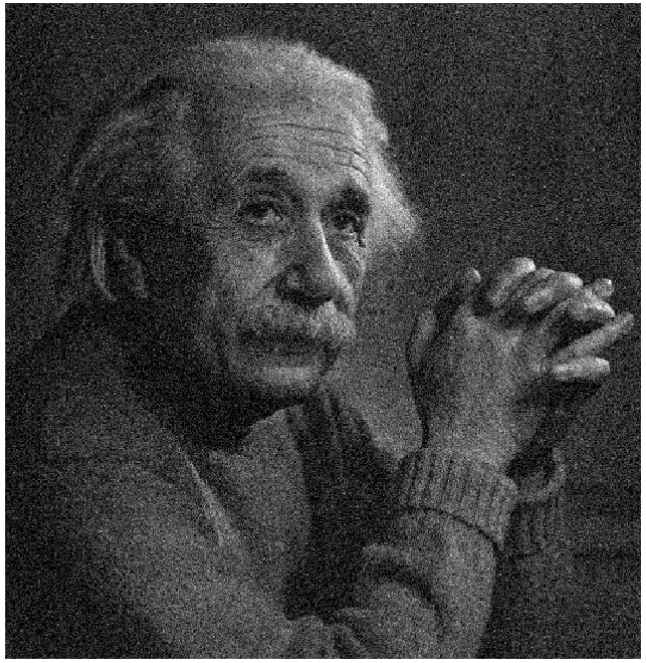




Slide credit Steve Seitz

Effects of noise



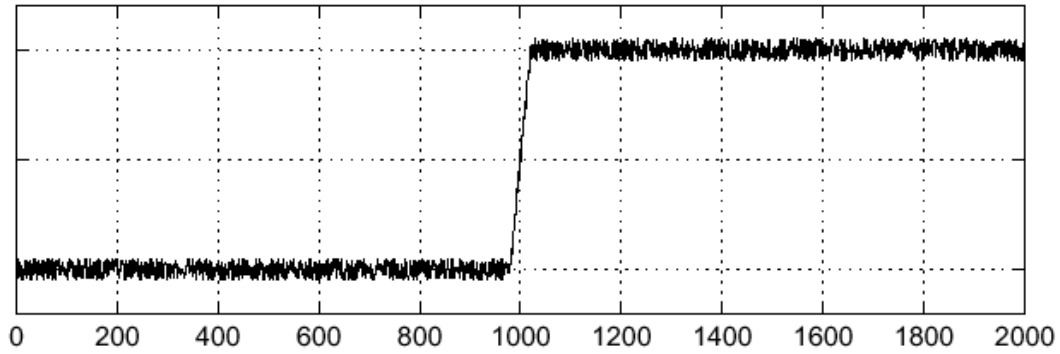


Effects of noise

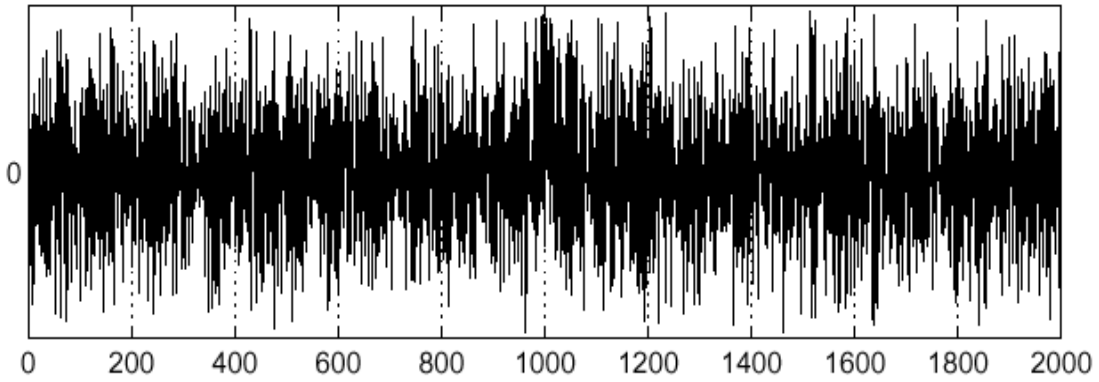
Consider a single row or column of the image

- Plotting intensity as a function of position gives a signal

$f(x)$



$\frac{d}{dx}f(x)$



Where is the edge?

Slide credit Steve Seitz

Effects of noise

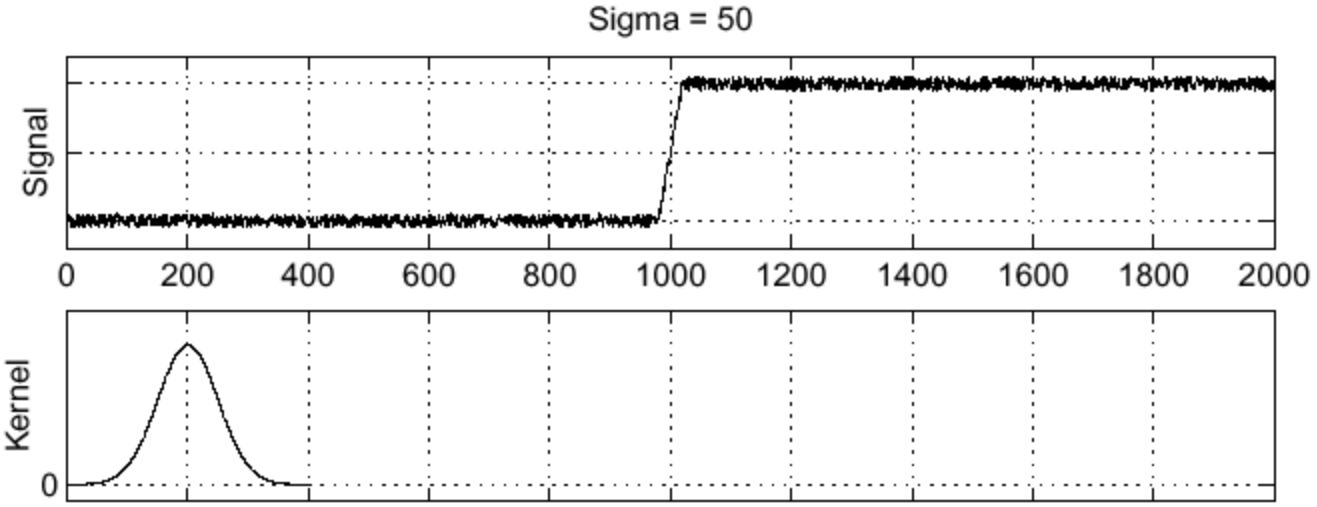
- Difference filters respond strongly to noise
  - Image noise results in pixels that look very different from their neighbors
  - Generally, the larger the noise the stronger the response
- What can we do about it?

Source: D. Forsyth

Solution: smooth first

f

h



Where is the edge?

Look for peaks in $\frac{\partial}{\partial x}(h \star f)$

Derivative theorem of convolution

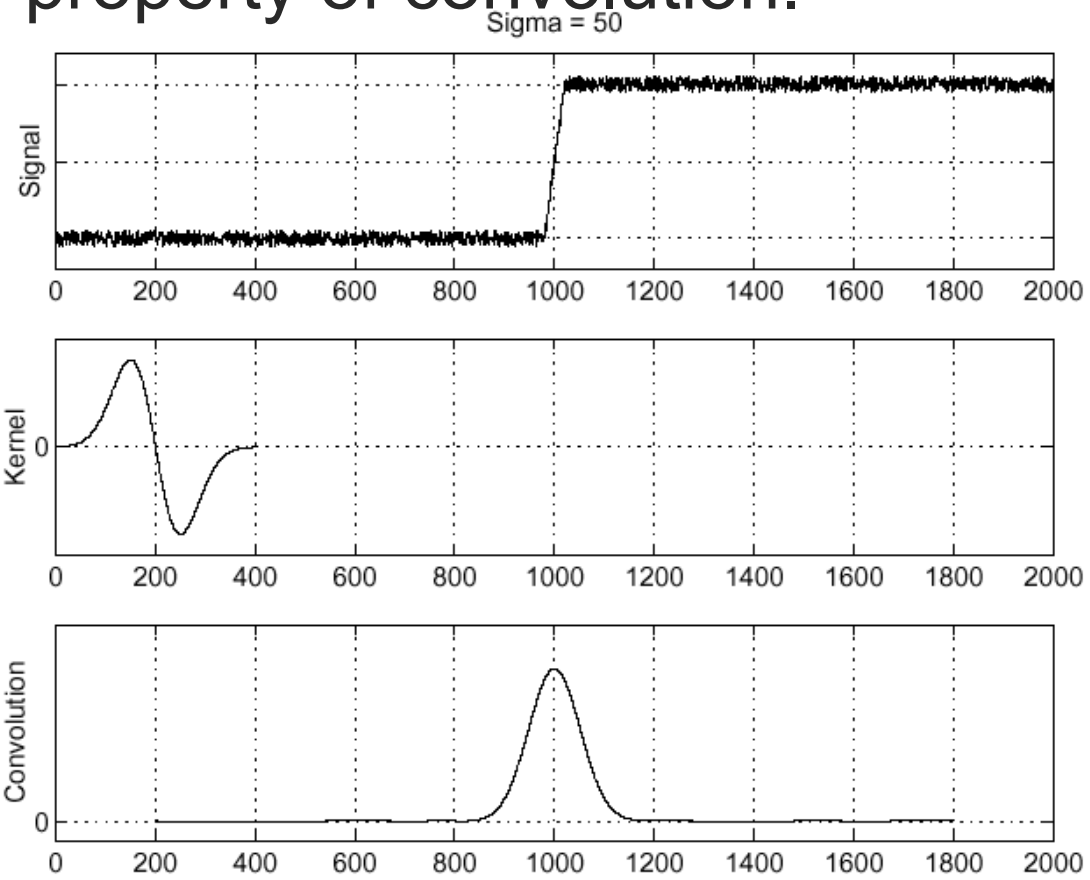
$$\frac{\partial}{\partial x}(h \star f) = (\frac{\partial}{\partial x}h) \star f$$

Differentiation property of convolution.

f

$\frac{\partial}{\partial x}h$

$(\frac{\partial}{\partial x}h) \star f$

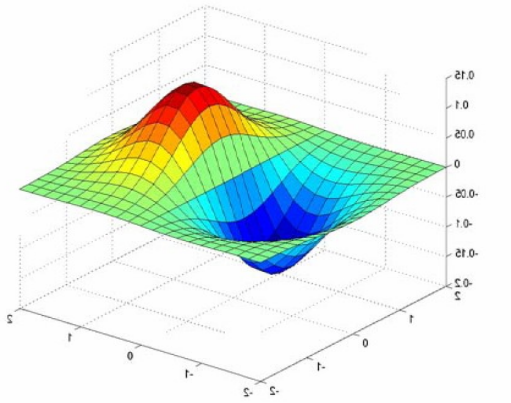


Slide credit Steve Seitz

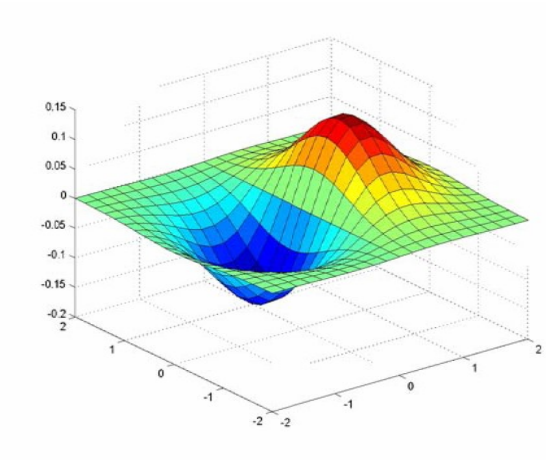
Derivative of Gaussian filters

$$(I \otimes g) \otimes h = I \otimes (g \otimes h)$$

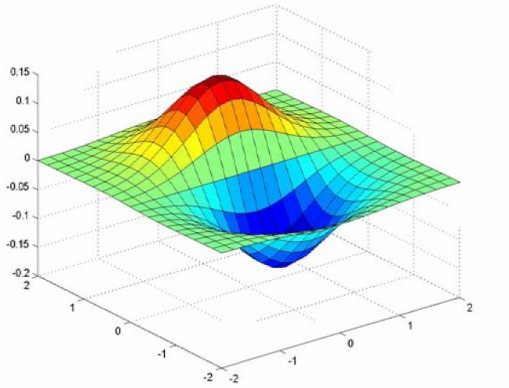
$$\begin{bmatrix} 0.0030 & 0.0133 & 0.0219 & 0.0133 & 0.0030 \\ 0.0133 & 0.0596 & 0.0983 & 0.0596 & 0.0133 \\ 0.0219 & 0.0983 & 0.1621 & 0.0983 & 0.0219 \\ 0.0133 & 0.0596 & 0.0983 & 0.0596 & 0.0133 \\ 0.0030 & 0.0133 & 0.0219 & 0.0133 & 0.0030 \end{bmatrix} \otimes \begin{bmatrix} 1 & -1 \end{bmatrix}$$



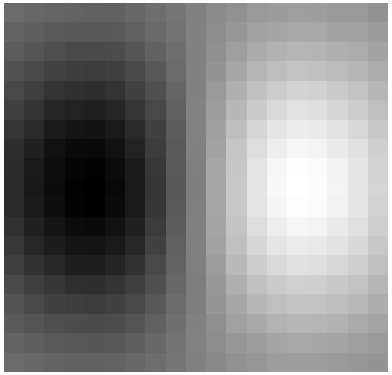
Derivative of Gaussian filters

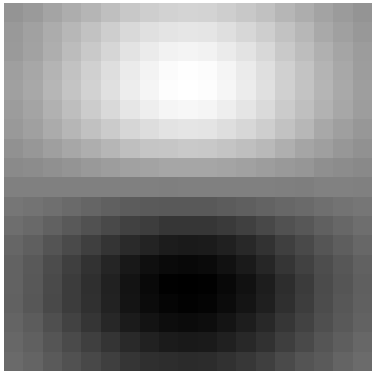


***x*-direction**



***y*-direction**





Source: L. Lazebnik

The Sobel Operator: A common approximation of derivative of gaussian

- Common approximation of derivative of Gaussian

$$\frac{1}{8}\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} \qquad s_x$$

$$\frac{1}{8}\begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix} \qquad s_y$$

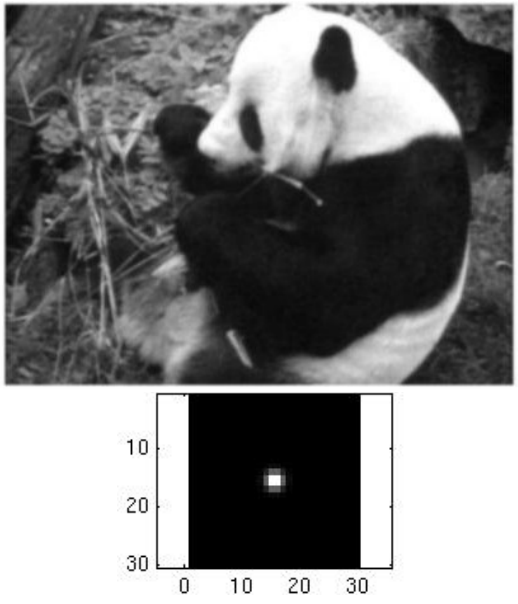
- The standard defn. of the Sobel operator omits the 1/8 term
  - doesn't make a difference for edge detection
  - the 1/8 term **is** needed to get the right gradient value

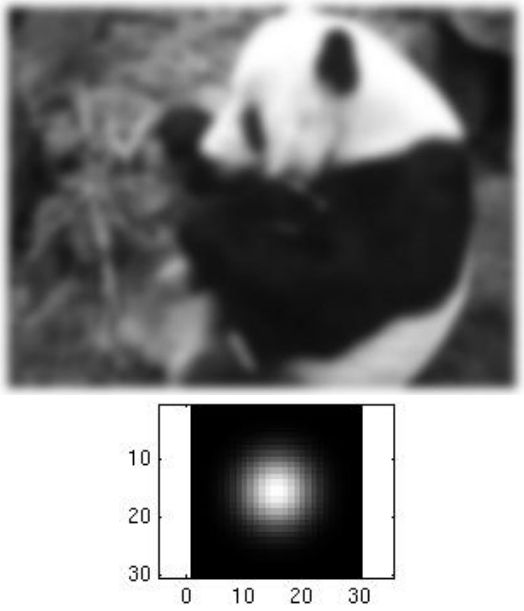
Mask properties

- Smoothing
  - Values positive
  - Sum to 1 → constant regions same as input
  - Amount of smoothing proportional to mask size
  - Remove “high-frequency” components; “low-pass” filter
- Derivatives
  - _____ signs used to get high response in regions of high contrast
  - Sum to ___ → no response in constant regions
  - High absolute value at points of high contrast

Smoothing with a Gaussian

Recall: parameter σ is the “scale” / “width” / “spread” of the Gaussian kernel, and controls the amount of smoothing.





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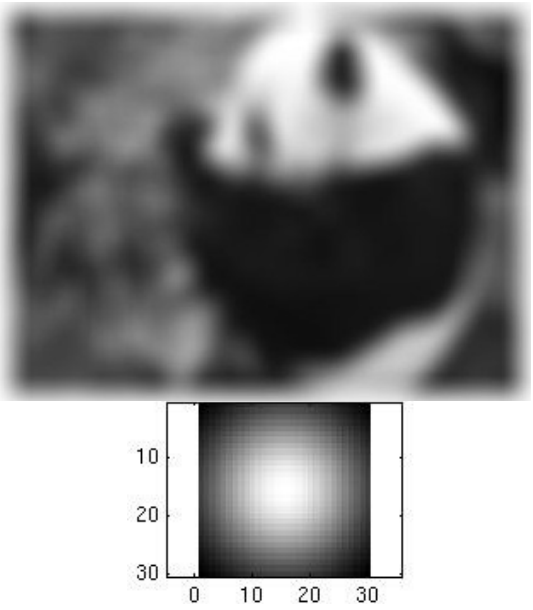


Effect of σ on derivatives

σ = 1 pixel

σ = 3 pixels

The apparent structures differ depending on Gaussian's scale parameter.
Larger values: larger scale edges detected
Smaller values: finer features detected

Slide credit: Kristen Grauman

Gradients -> edges

Primary edge detection steps:

1. Smoothing: suppress noise
2. Edge enhancement: filter for contrast
3. Edge localization

   Determine which local maxima from filter output are actually edges vs. noise

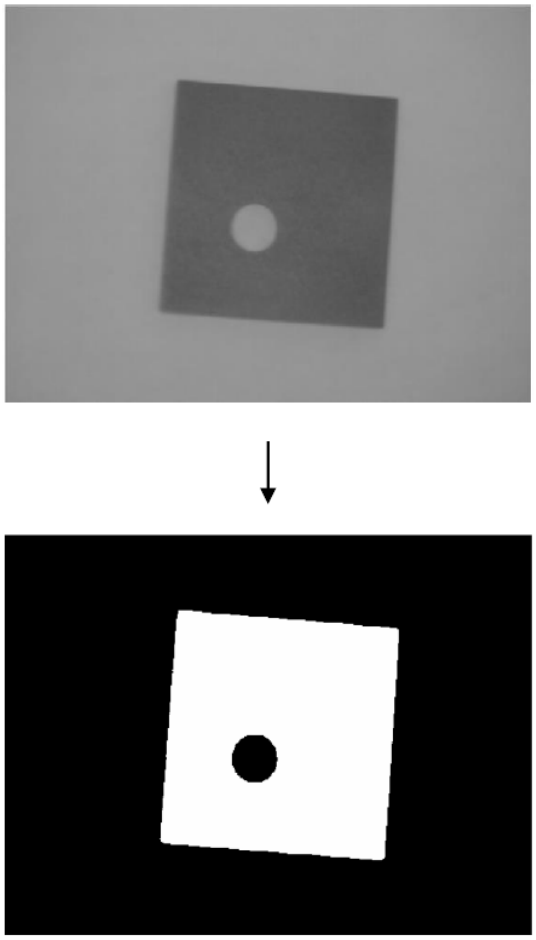
   Thresholding and thinning

$\Rightarrow$ Larger values: larger-scale edges detected

$\Rightarrow$ Smaller values: finer features detected

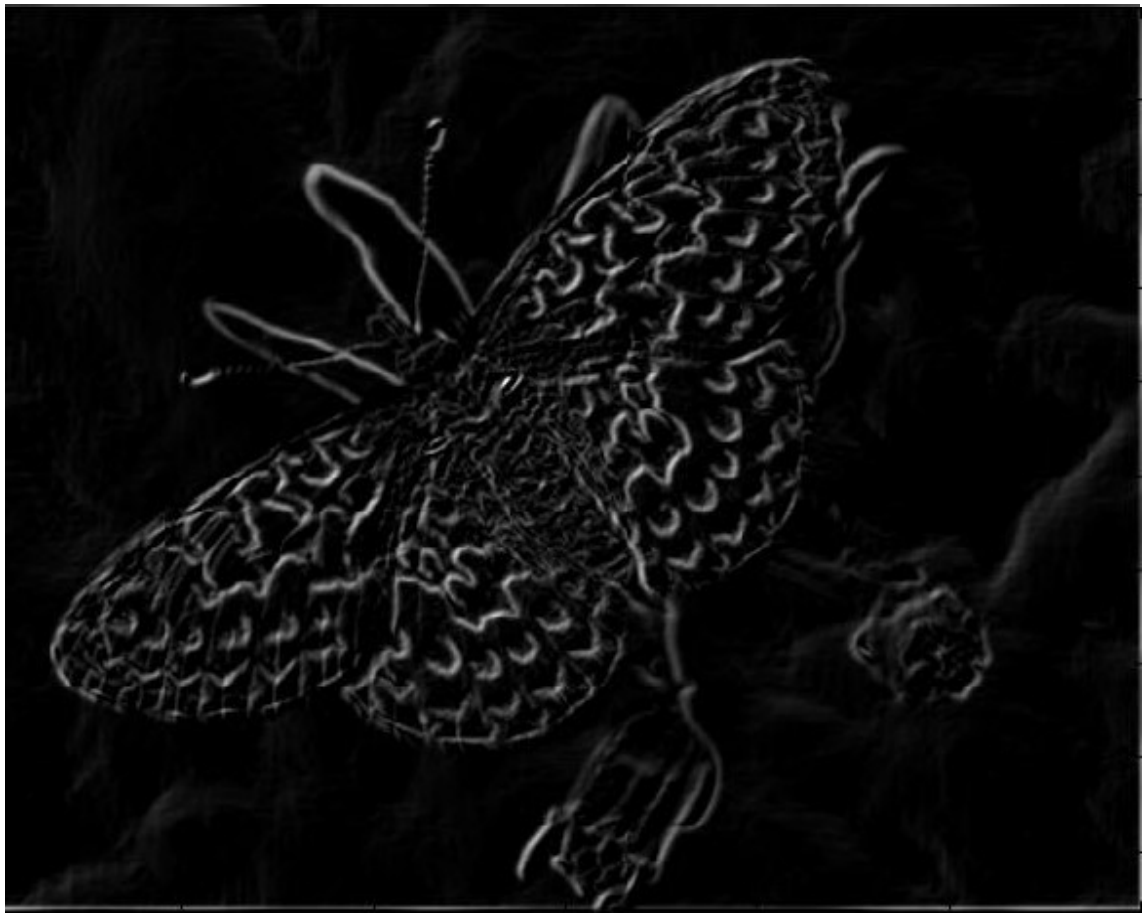
Thresholding

- Choose a threshold value t
- Set any pixels less than t to zero (off)
- Set any pixels greater than or equal to t to one (on)



Original image

Gradient magnitude image



Lower threshold

Higher threshold

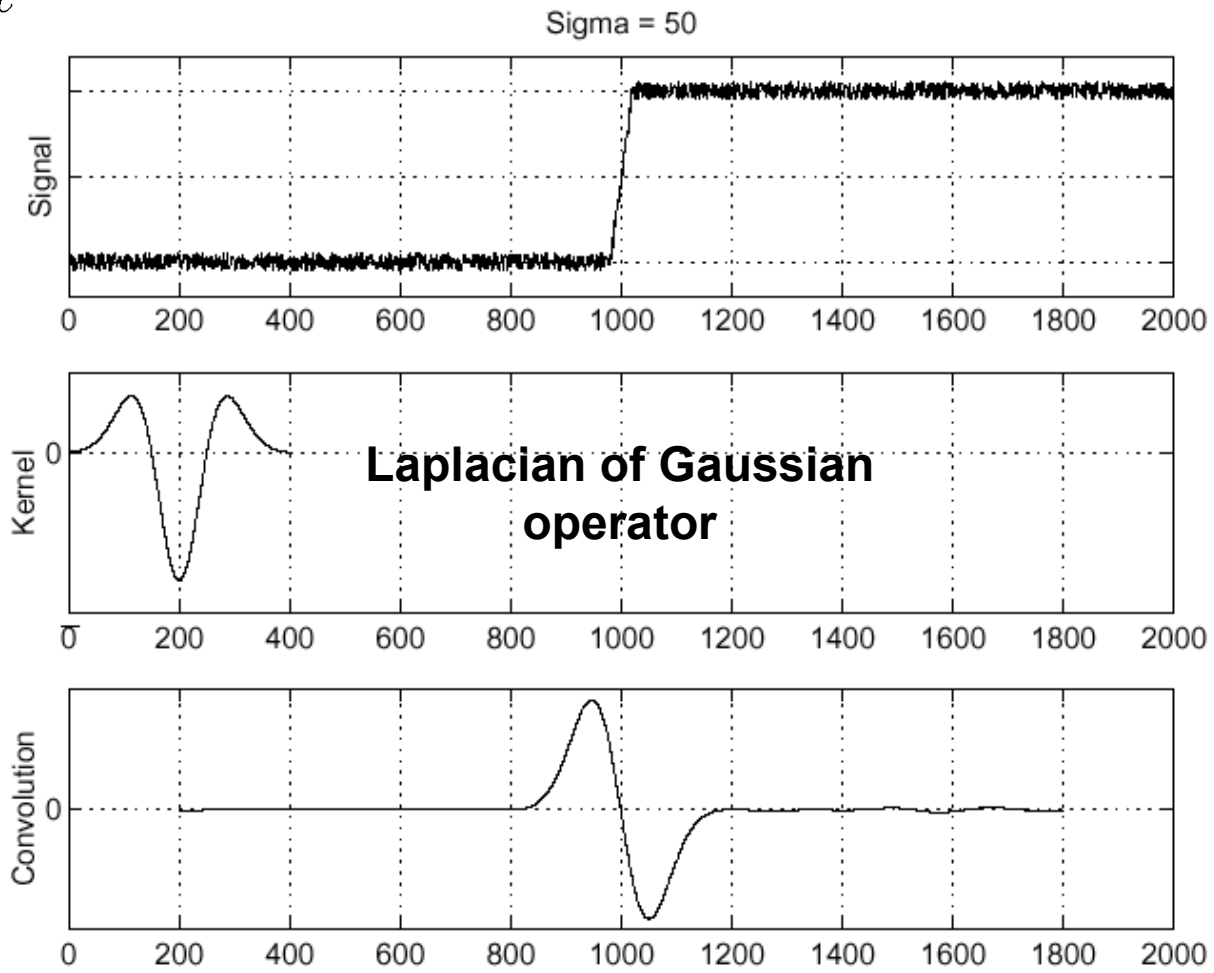
Laplacian of Gaussian

Consider $\frac{\partial^2}{\partial x^2}(h \star f)$

f

$\frac{\partial^2}{\partial x^2} h$

$(\frac{\partial^2}{\partial x^2} h) \star f$

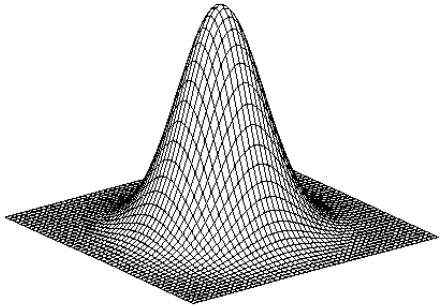


Where is the edge?

Zero-crossings of bottom graph

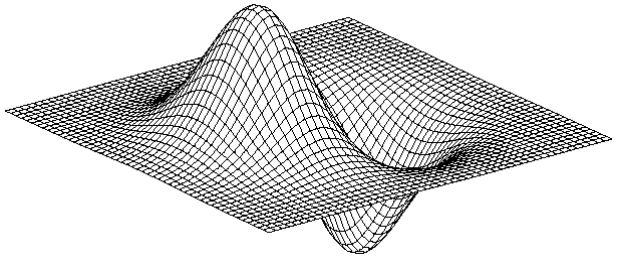
Slide credit: Steve S

2D edge detection filters



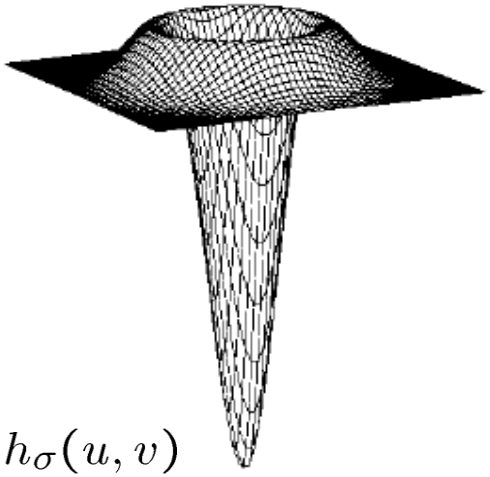
Gaussian

$$h_\sigma(u,v) = \frac{1}{2\pi\sigma^2} e^{-\frac{u^2+v^2}{2\sigma^2}}$$



derivative of Gaussian

$$\frac{\partial}{\partial x} h_\sigma(u,v)$$



$\nabla^2 h_\sigma(u,v)$

Laplacian of Gaussian

- ∇^2 is the Laplacian operator:

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

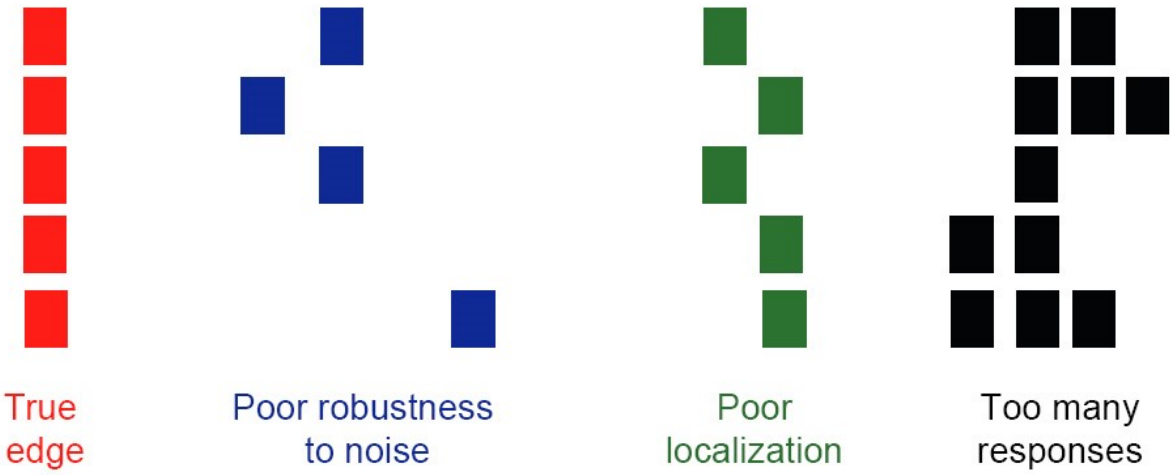
Slide credit: Steve Seitz

Today's class

- Introduction to edge detection
- Gradients and edges
- **Canny edge detector**
- Object contour
- Pb edge detector
- Straight line detection

Designing an edge detector

- Criteria for an “optimal” edge detector:
  - **Good detection:** minimize the probability of false positives (detecting spurious edges caused by noise), as well as that of false negatives (missing real edges)
  - **Good localization:** must be as close as possible to the true edges
  - **Single response:** the detector must return one point only for each true edge point;



Basic Comparisons of Edge Operators

Gradient:

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right]$$

Good Localization
Noise Sensitive
Poor Detection

Roberts (2 x 2):

0	1
-1	0

1	0
0	-1

Sobel (3 x 3):

-1	0	1
-1	0	1
-1	0	1

1	1	1
0	0	0
-1	-1	1

Sobel (5 x 5):

-1	-2	0	2	1
-2	-3	0	3	2
-3	-5	0	5	3
-2	-3	0	3	2
-1	-2	0	2	1

1	2	3	2	1
2	3	5	3	2
0	0	0	0	0
-2	-3	-5	-3	-2
-1	-2	-3	-2	-1

Poor Localization
Less Noise Sensitive
Good Detection

Canny edge detector

- This is probably the most widely used edge detector in computer vision
- Theoretical model: step-edges corrupted by additive Gaussian noise
- Canny has shown that the first derivative of the Gaussian closely approximates the operator that optimizes the product of *signal-to-noise ratio* and localization

J. Canny, ***A Computational Approach To Edge Detection***, IEEE Trans. Pattern Analysis and Machine Intelligence, 8:679-714, 1986.

Canny edge detector

- Filter image with derivative of Gaussian
- Find magnitude and orientation of gradient
- **Non-maximum suppression**:
  - Thin wide “ridges” down to single pixel width
- **Linking and thresholding** (**hysteresis**):
  - Define two thresholds: low and high
  - Use the high threshold to start edge curves and the low threshold to continue them
- MATLAB: `edge(image, 'canny');`
- `>>help edge`

Source: D. Lowe, L. Fei-Fei

The Canny edge detector

original image (Lena)

Slide credit: Steve S

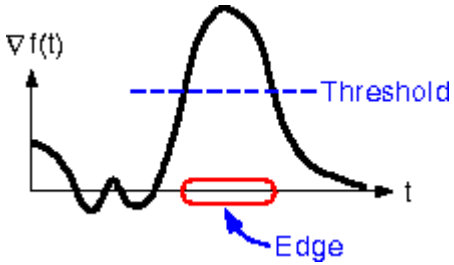
The Canny edge detector

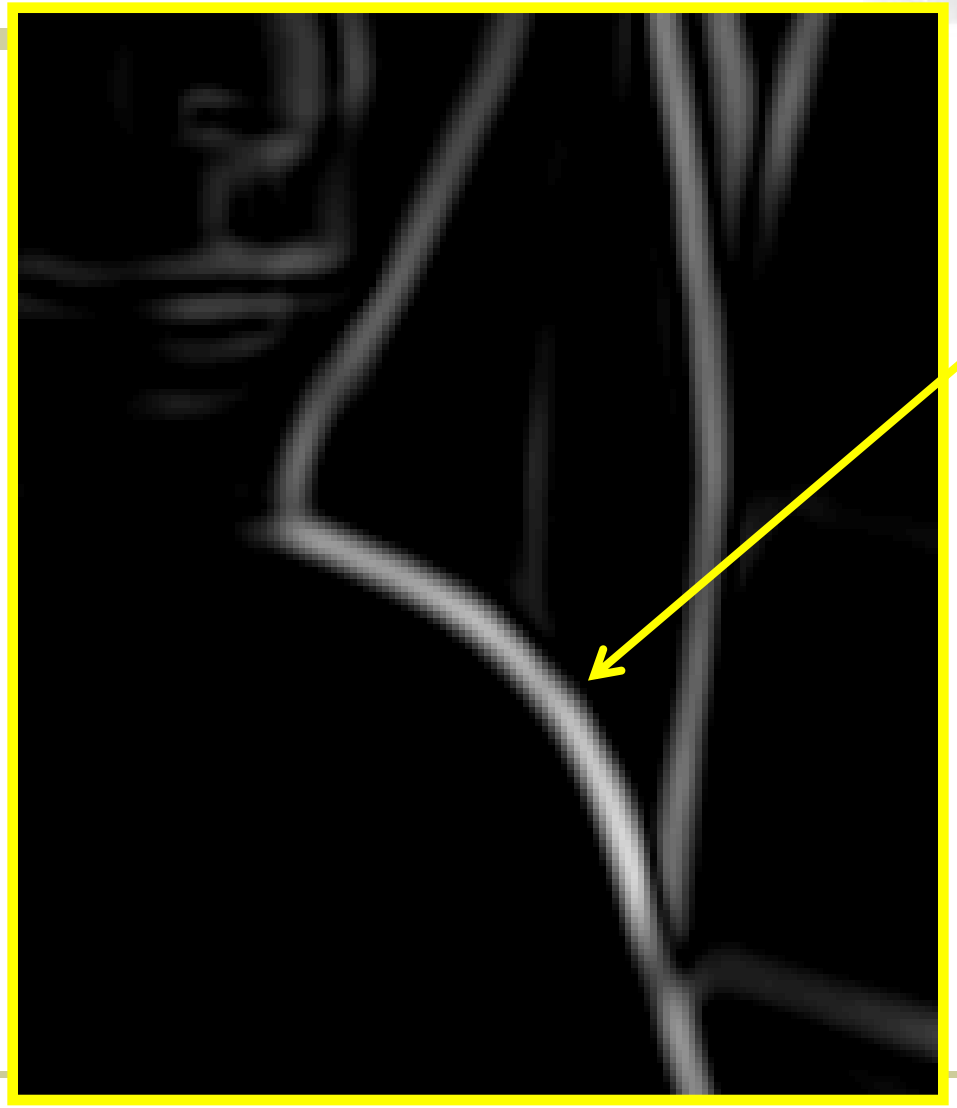
norm of the gradient

The Canny edge detector

thresholding

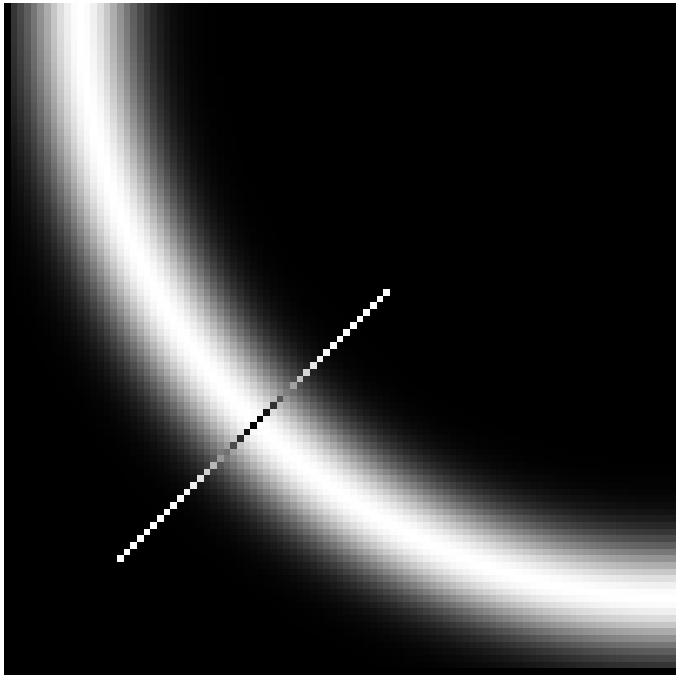
The Canny edge detector

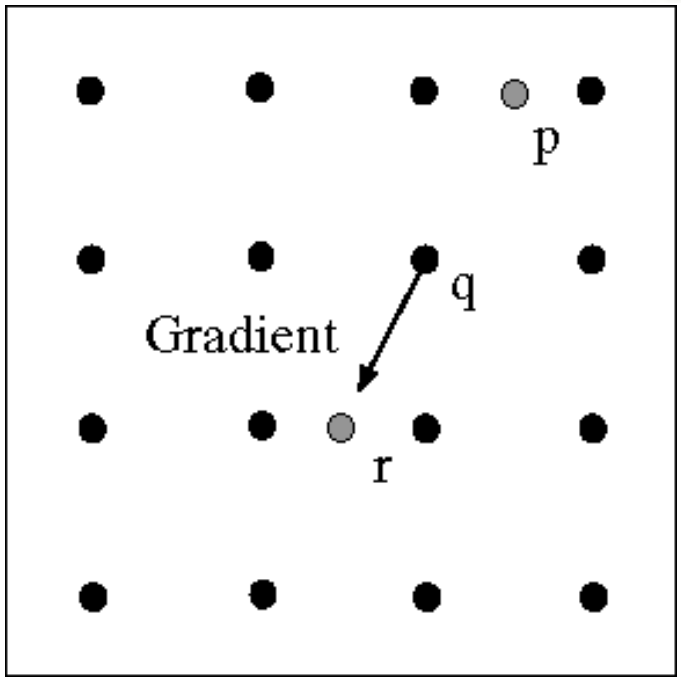




How to turn these thick regions of the gradient into curves?

Non-maximum suppression



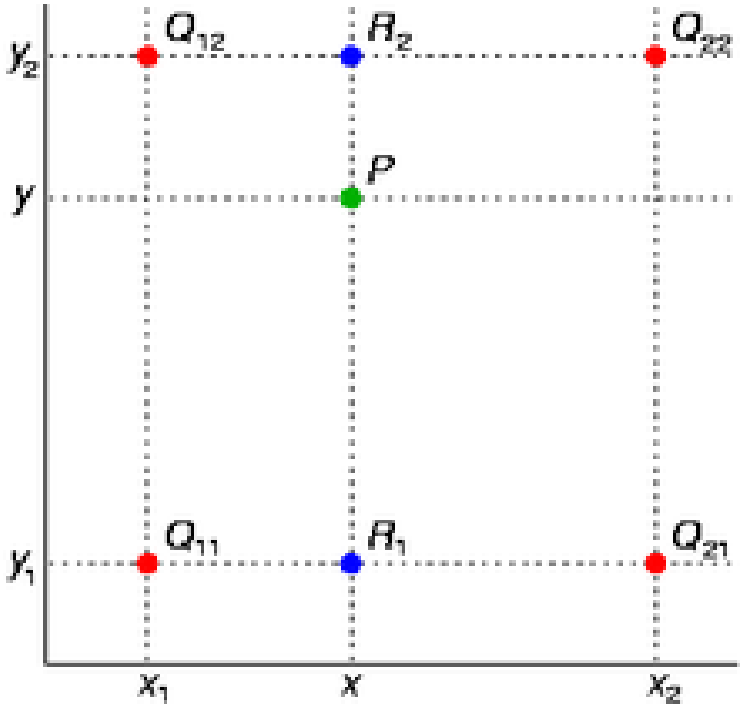


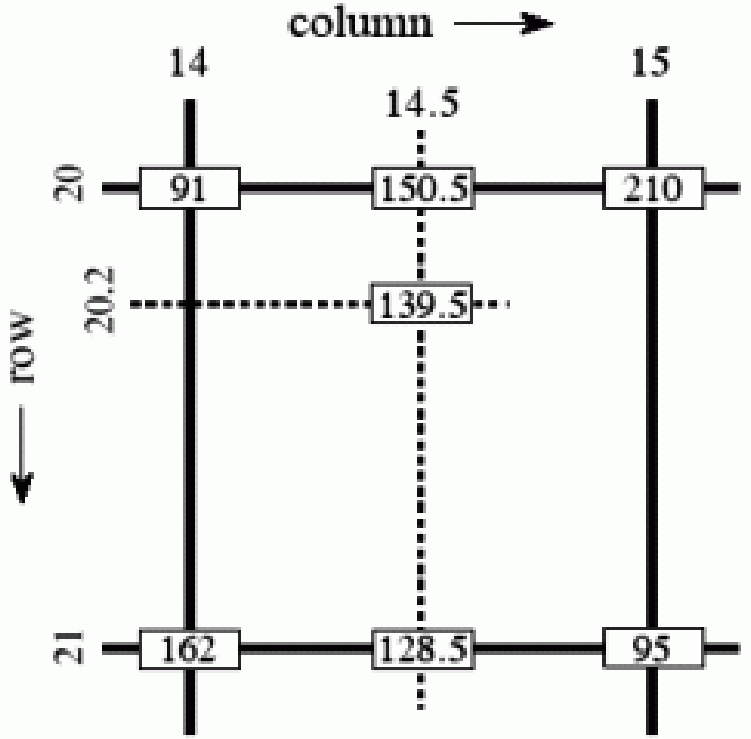
Check if pixel is local maximum along gradient direction, select single max across width of the edge

- requires checking interpolated pixels p and r

Bilinear Interpolation

$$f(x,y) \approx \begin{bmatrix} 1-x & x \end{bmatrix} \begin{bmatrix} f(0,0) & f(0,1) \\ f(1,0) & f(1,1) \end{bmatrix} \begin{bmatrix} 1-y \\ y \end{bmatrix}.$$





http://en.wikipedia.org/wiki/Bilinear_interpolation

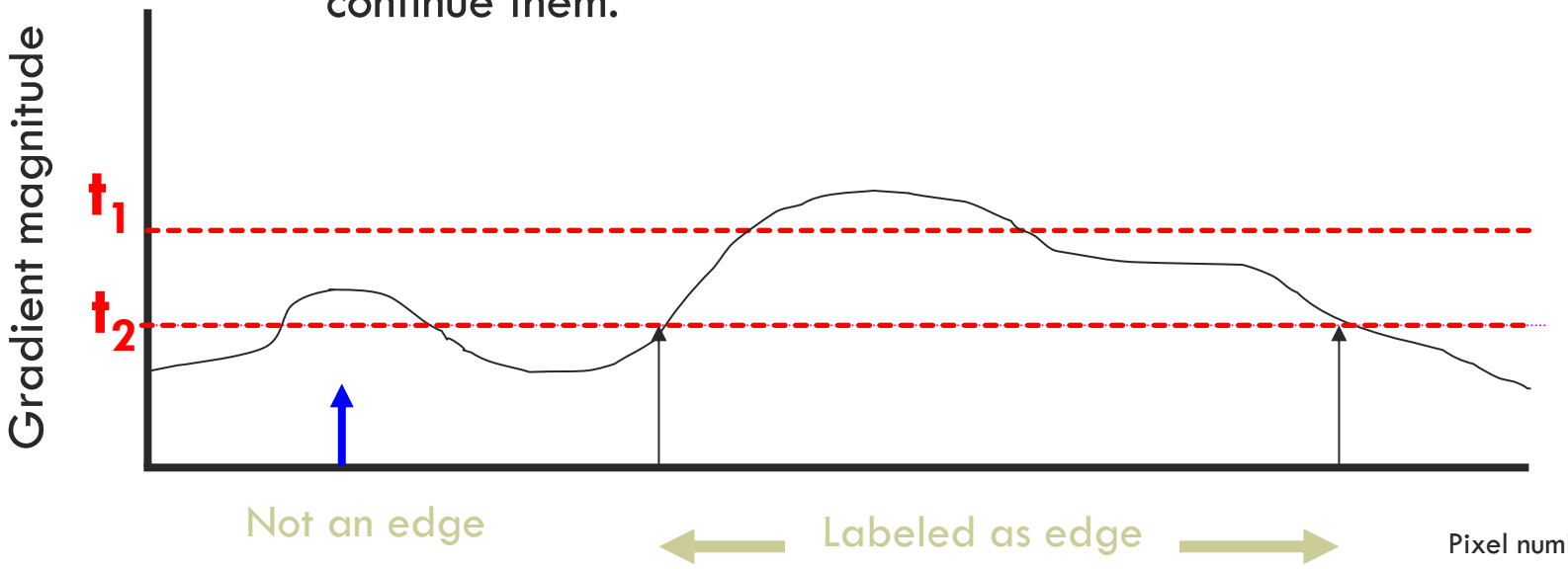
Before non-max suppression

After non-max suppression

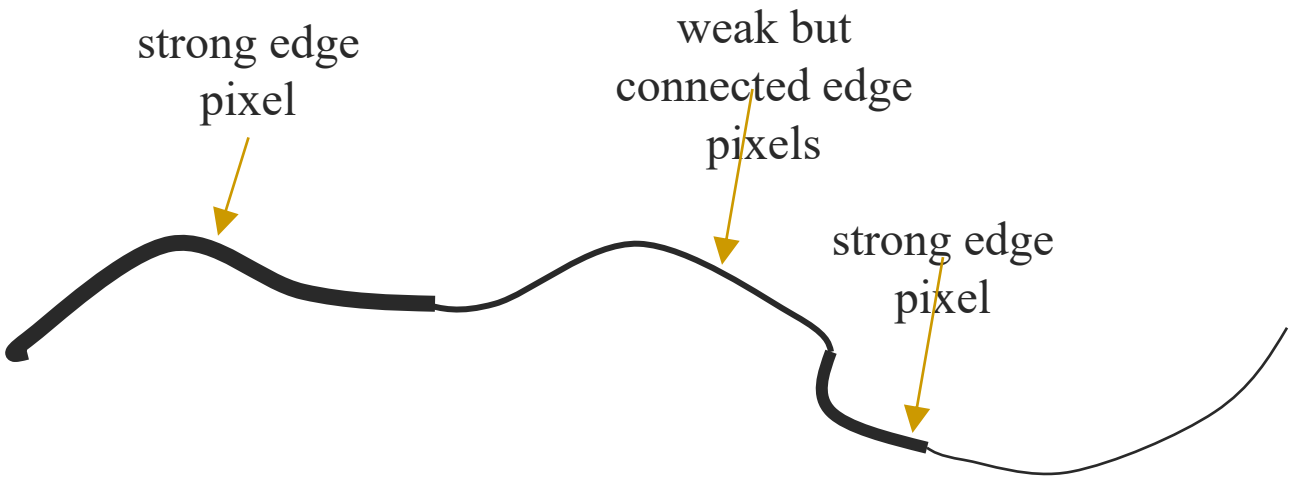
Problem: pixels along this edge didn't survive the thresholding

Closing edge gaps

- Check that maximum value of gradient value is sufficiently large
  - use **hysteresis**
    - use a high threshold to start edge curves and a low threshold to continue them.



Hysteresis thresholding



Source: S. Seitz

Hysteresis thresholding

- Threshold at low/high levels to get weak/strong edge pixels
- Do connected components, starting from strong edge pixels

Final Canny Edges

Effect of σ (Gaussian kernel spread/size)

original

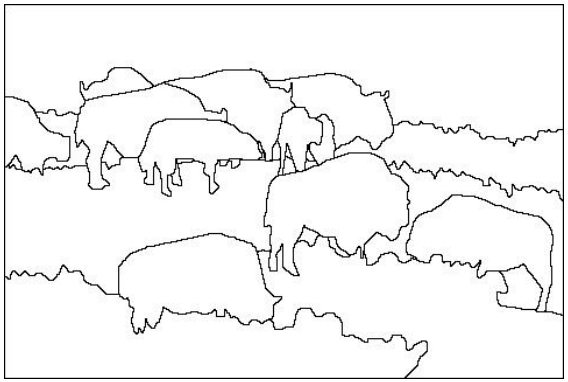
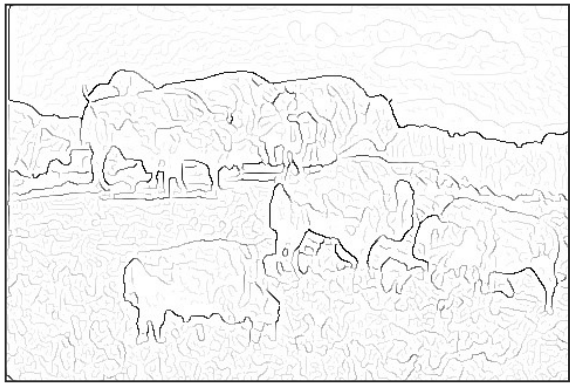
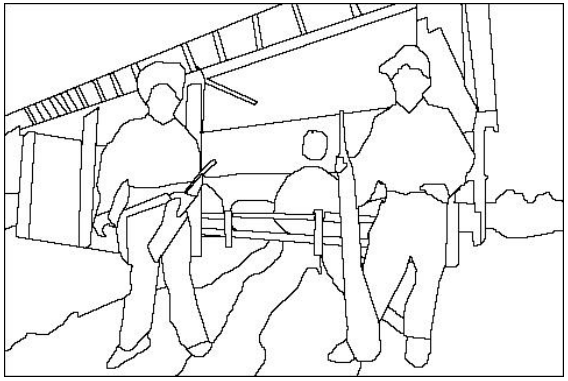
Canny with $\sigma = 1$

Canny with $\sigma = 2$

The choice of σ depends on desired behavior

- large σ detects large scale edges
- small σ detects fine features

Low-level edges vs. perceived contours

image	human segmentation	gradient magnitude
		
		

- Berkeley segmentation database:
  http://www.eecs.berkeley.edu/Research/Projects/CS/vision/grouping/segbench/

A Database of Human Segmented Natural Images and its Application to Evaluating Segmentation Algorithms and Measuring Ecological Statistics

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This paper presents a database containing 'ground truth' segmentations produced by humans for images of a wide variety of natural scenes. We define an error measure which quantifies the consistency between segmentations of differing granularities and find that different human segmentations of the same image are highly consistent. Use of this dataset is demonstrated in two applications: (1) evaluating the performance of segmentation algorithms and (2) measuring probability distributions associated with Gestalt grouping factors as well as statistics of image region properties.

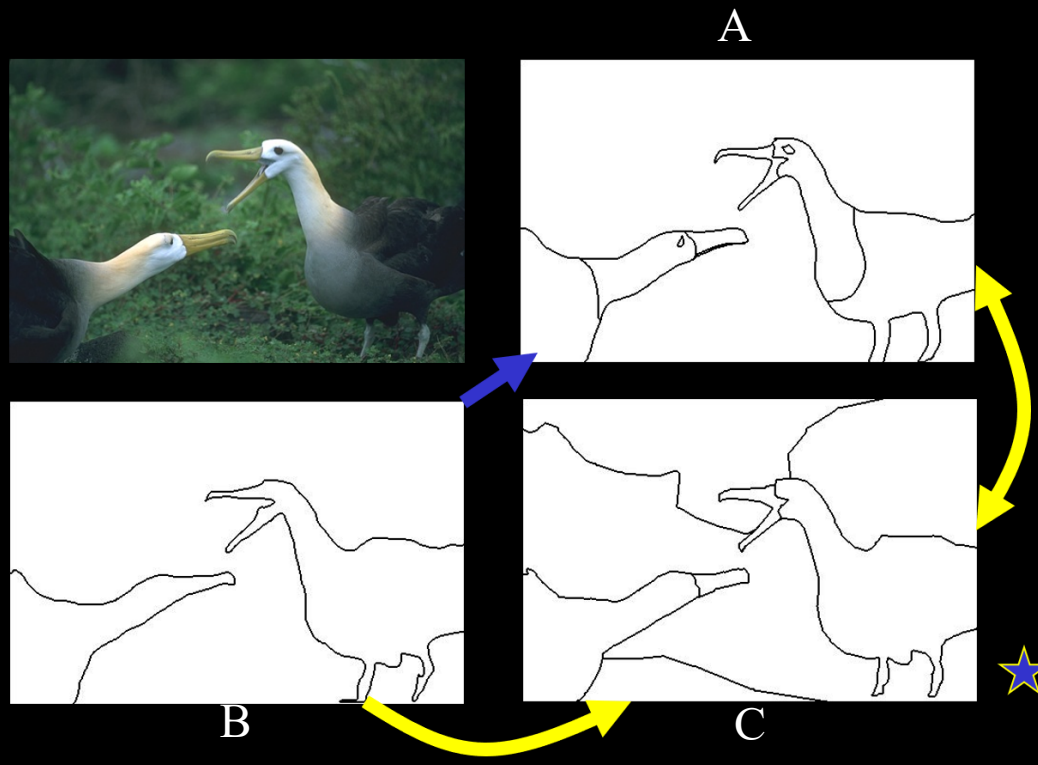
A database of human segmented natural images and its application to evaluating segmentation algorithms and measuring ecological statistics
Proceedings Eighth IEEE International Conference on …, 2001 - ieeexplore.ieee.org
This paper presents a database containing'ground truth'segmentations produced by humans for images of a wide variety of natural scenes. We define an error measure which quantifies …
☆ Notes 99 References Cited by 9120 Related articles

Protocol

You will be presented a photographic image. Divide the image into some number of segments, where the segments represent "things" or "parts of things" in the scene. The number of segments is up to you, as it depends on the image. Something between 2 and 30 is likely to be appropriate. It is important that all of the segments have approximately equal importance.

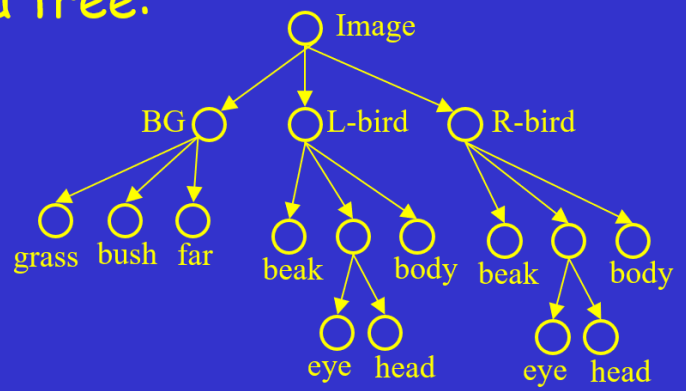
- Custom segmentation tool
- Subjects obtained from work-study program (UC Berkeley undergraduates)

Consistency



- A,C are refinements of B
- A,C are mutual refinements
- A,B,C represent the same percept
  - Attention accounts for differences

Perceptual organization forms a tree:



☆ Two segmentations are consistent when they can be explained by the same segmentation tree (i.e. they could be derived from a single perceptual organization).

Dataset Summary

- 30 subjects, age 19-23
  - 17 men, 13 women
  - 9 with artistic training
- 8 months
- 1,458 person hours
- 1,020 Corel images
- 11,595 Segmentations
  - 5,555 color, 5,554 gray, 486 inverted/negated

Today's class

- Introduction to edge detection
- Gradients and edges
- Canny edge detector
- Object contour
- **Pb edge detector**
- Straight line detection

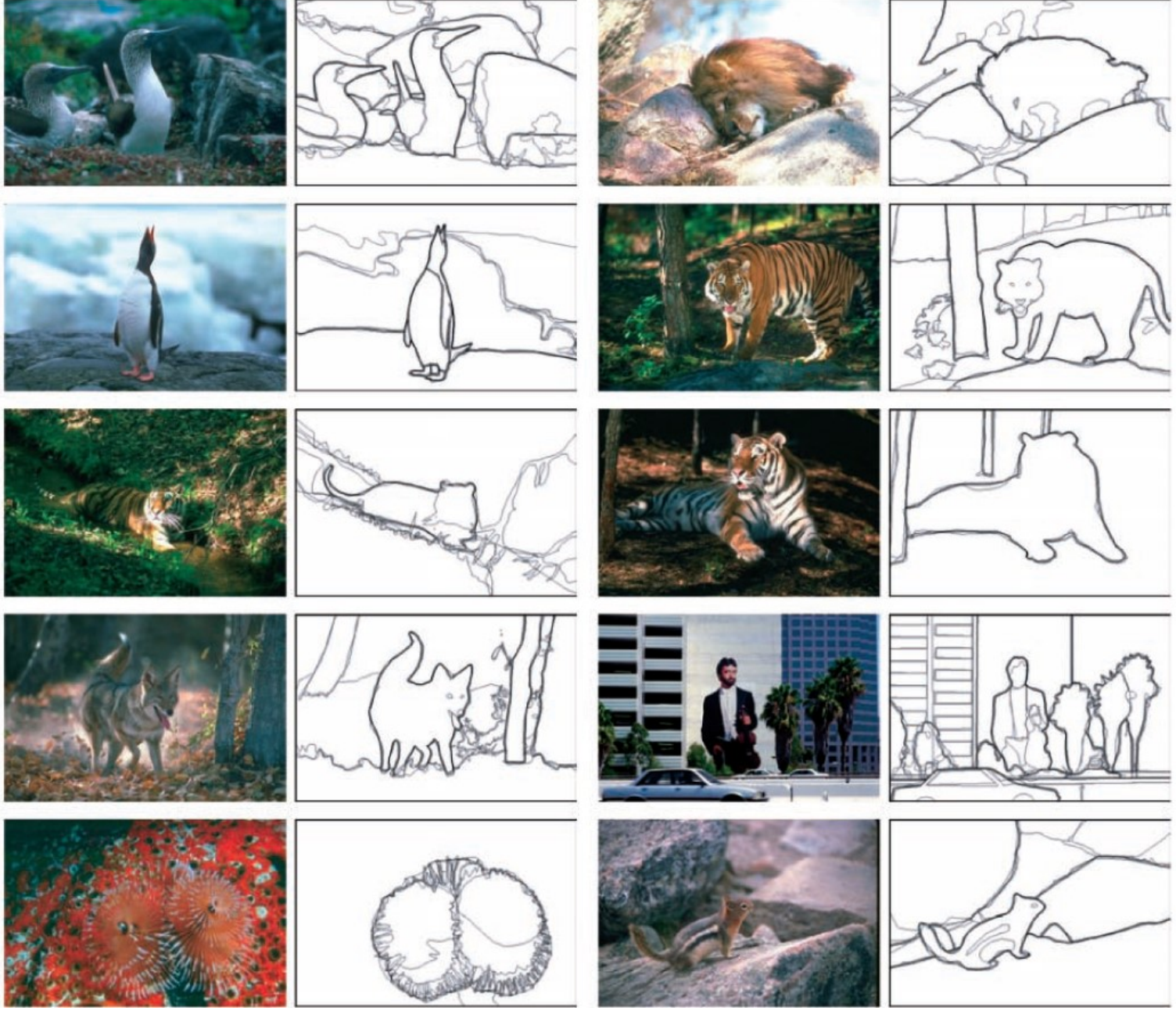
Learning to Detect Natural Image Boundaries Using Local Brightness, Color, and Texture Cues

David R. Martin, *Member*, *IEEE*, Charless C. Fowlkes, and Jitendra Malik, *Member*, *IEEE*

Abstract—The goal of this work is to accurately detect and localize boundaries in natural scenes using local image measurements. We formulate features that respond to characteristic changes in brightness, color, and texture associated with natural boundaries. In order to combine the information from these features in an optimal way, we train a classifier using human labeled images as ground truth. The output of this classifier provides the posterior probability of a boundary at each image location and orientation. We present precision-recall curves showing that the resulting detector significantly outperforms existing approaches. Our two main results are 1) that cue combination can be performed adequately with a simple linear model and 2) that a proper, explicit treatment of texture is required to detect boundaries in natural images.

Index Terms—Texture, supervised learning, cue combination, natural images, ground truth segmentation data set, boundary detection, boundary localization.

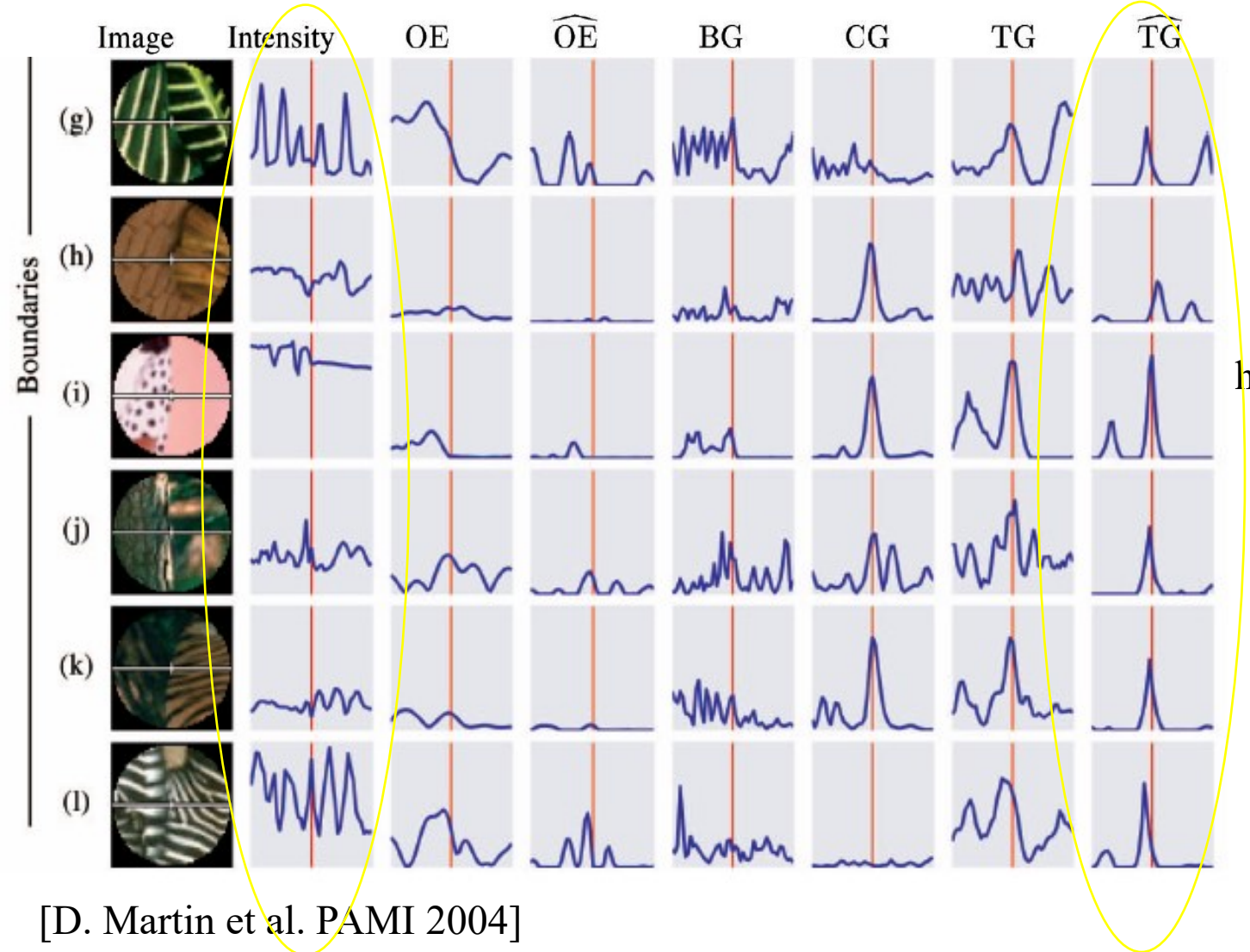
Learn from humans **which combination of features** is most indicative of a "good" contour?



[D. Martin et al. PAMI 2004]

Human-marked segment boundaries

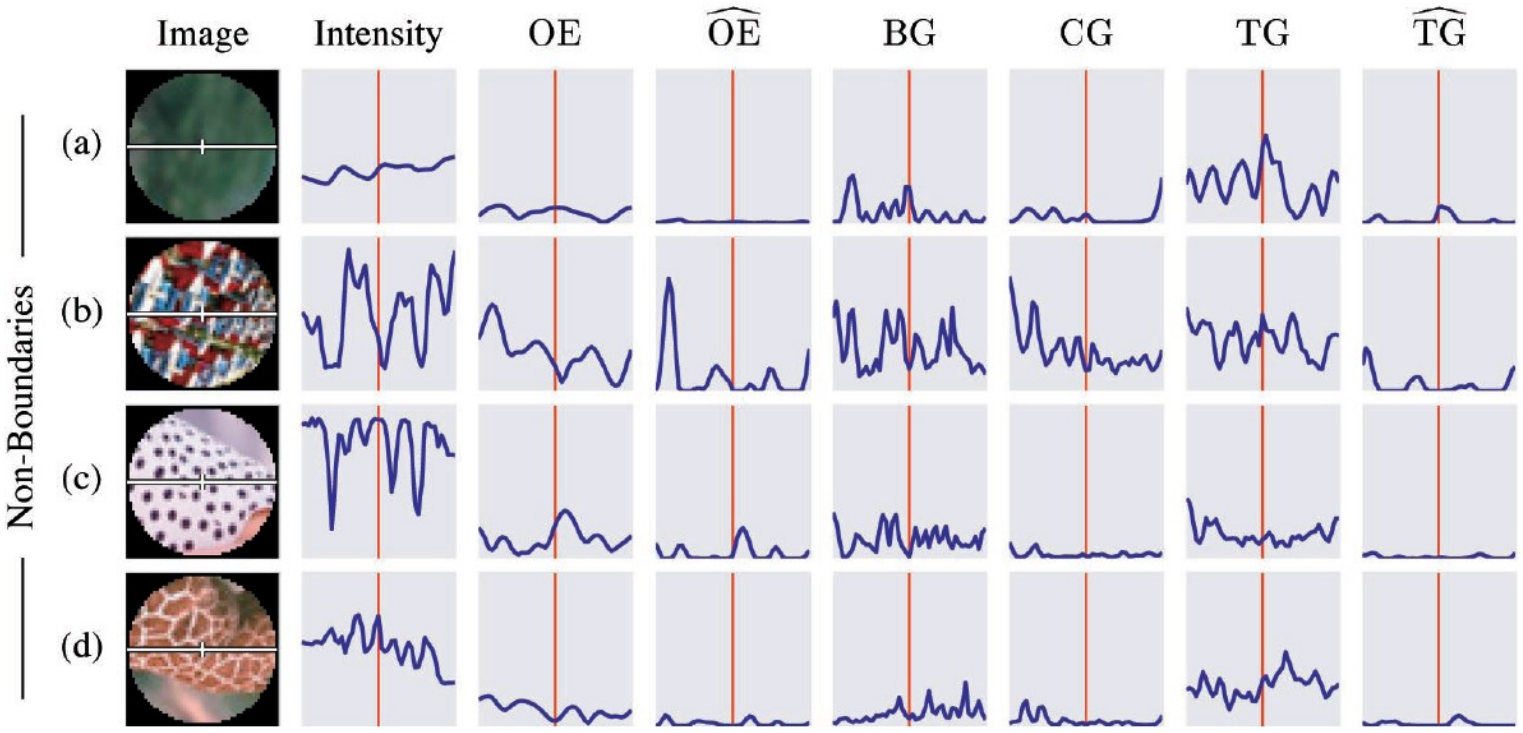
What features are responsible for perceived edges?



Feature profiles (oriented energy, brightness, color, and texture gradients) along the patch's horizontal diameter

[D. Martin et al. PAMI 2004]

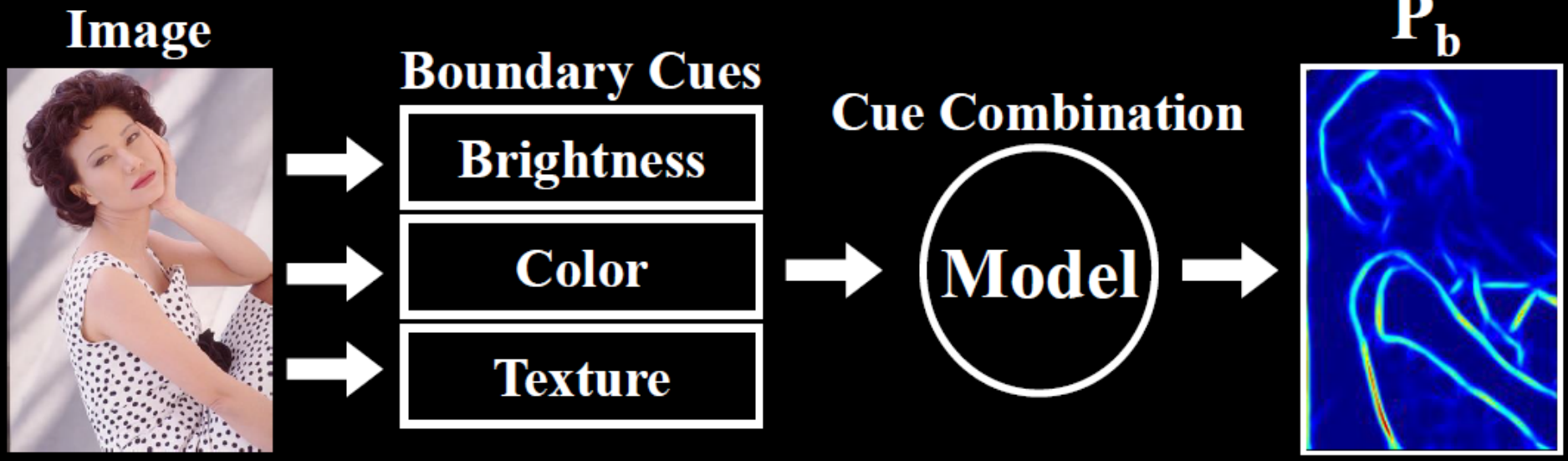
What features are responsible for perceived edges?



Feature profiles (oriented energy, brightness, color, and texture gradients) along the patch's horizontal diameter

[D. Martin et al. PAMI 2004]

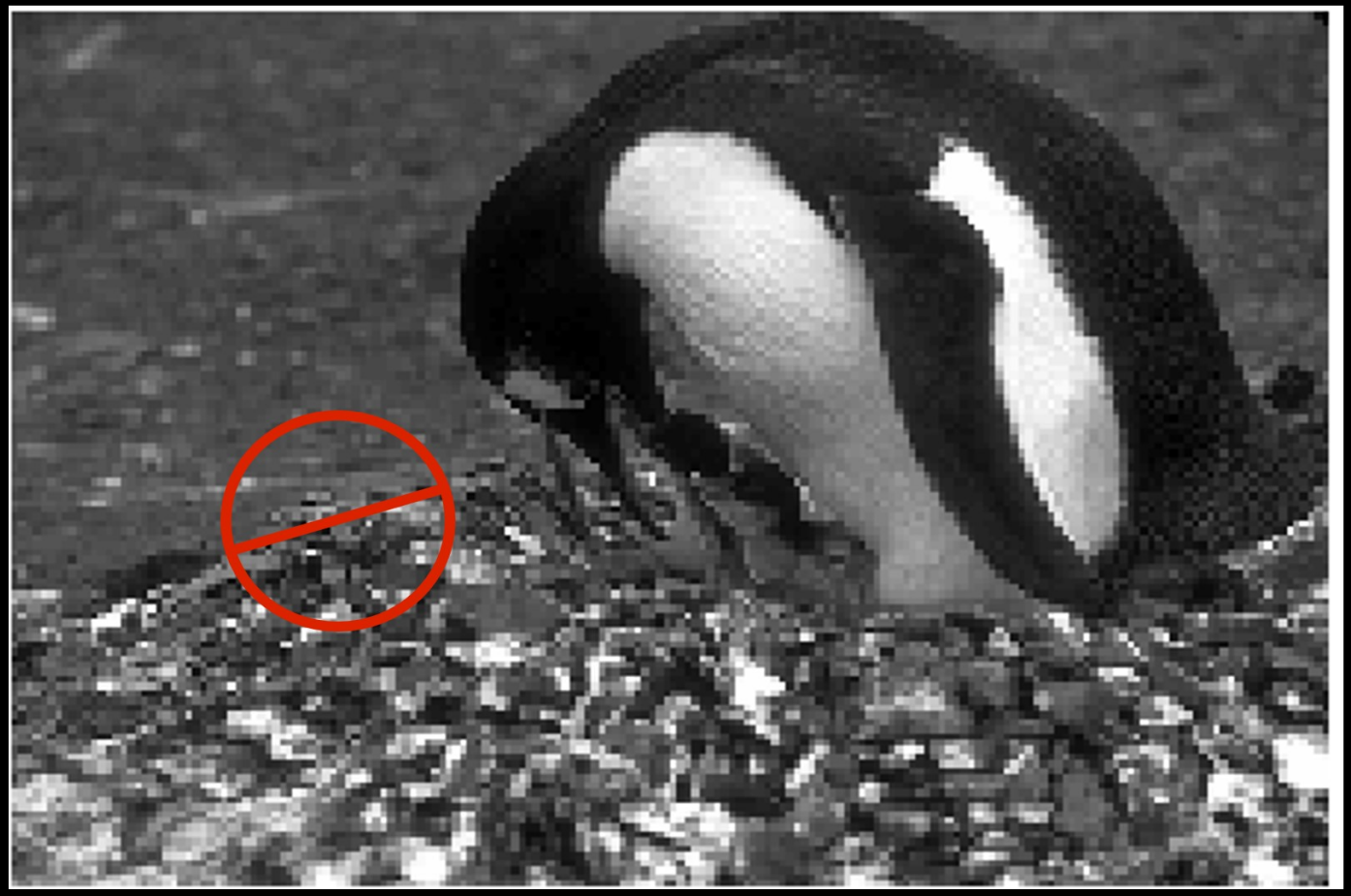
Dataflow



Challenges: texture cue, cue combination

Goal: learn the posterior probability of a boundary $P_b(x,y,\theta)$ from local information only

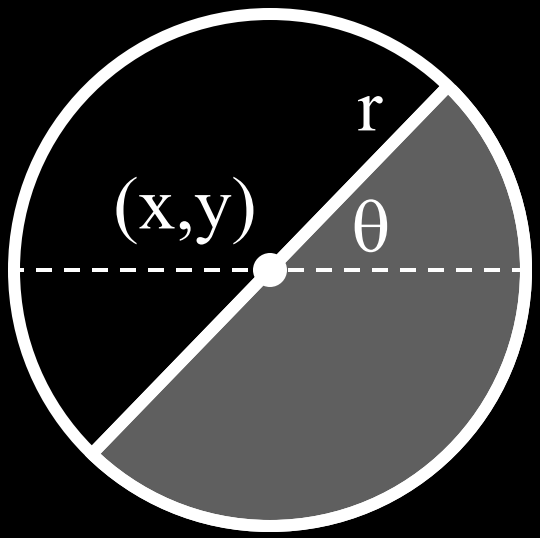
Oriented Feature Gradient



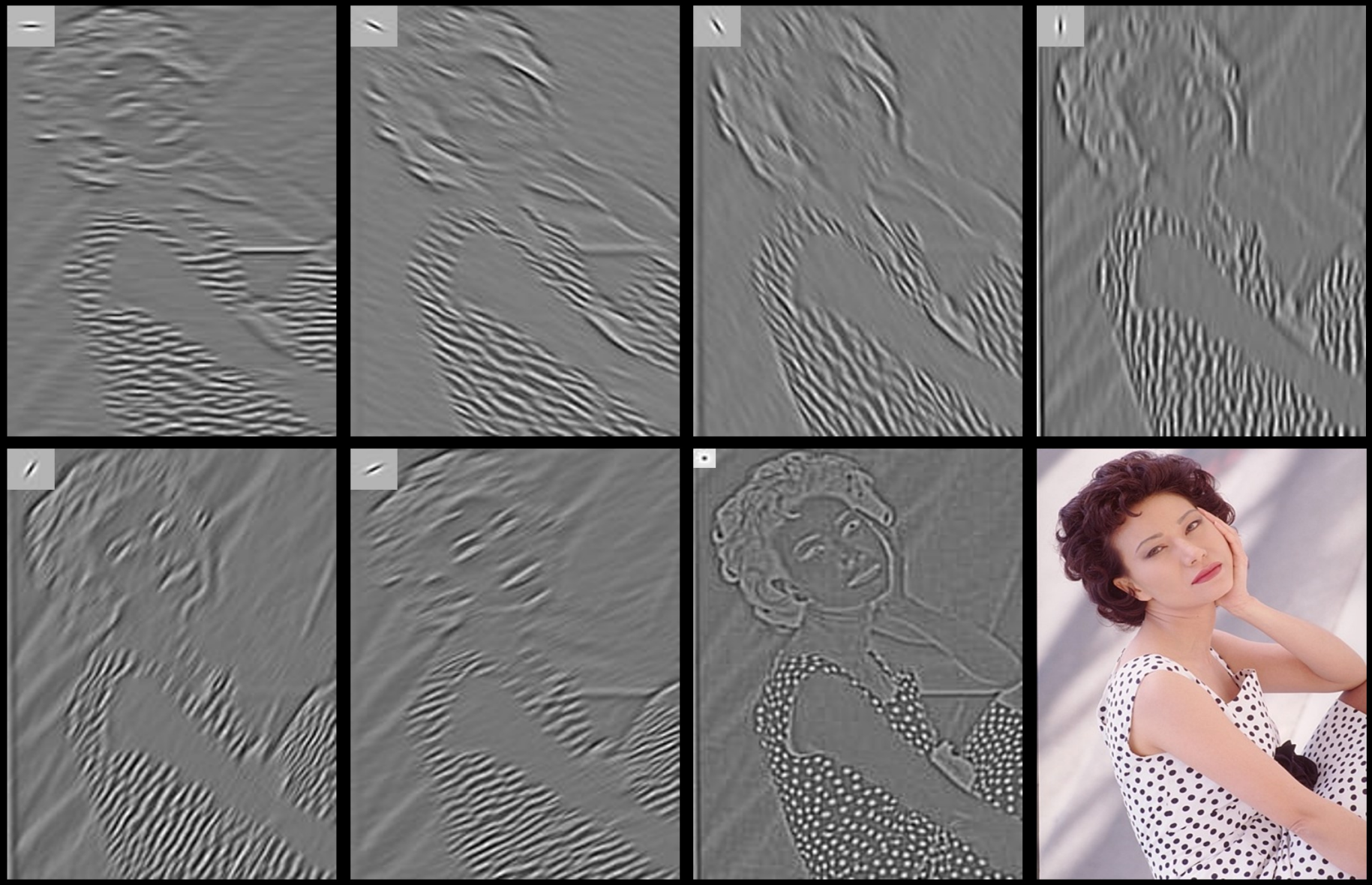
Brightness and Color Features

- 1976 CIE L*a*b* colorspace
- Brightness Gradient BG(x,y,r,θ)
  - χ^2 difference in L* distribution
- Color Gradient CG(x,y,r,θ)
  - χ^2 difference in a* and b* distributions

$$\chi^2(g,h) = \frac{1}{2}\sum_i \frac{(g_i - h_i)^2}{g_i + h_i}$$



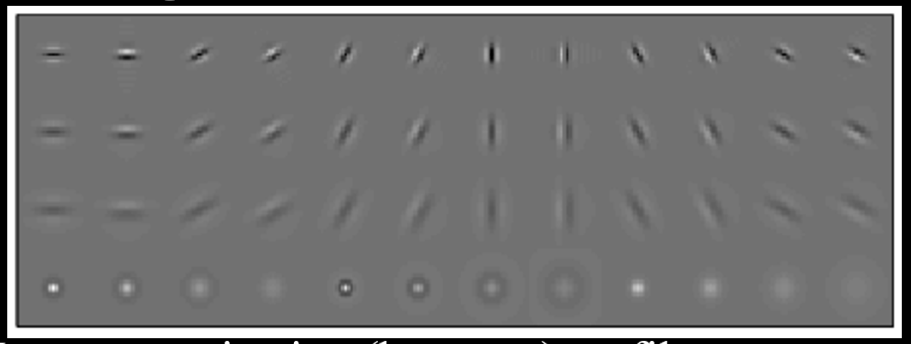
Filter Outputs



2D Textons

- Goal: find canonical local features in a texture;

  1) Filter image with linear filters:

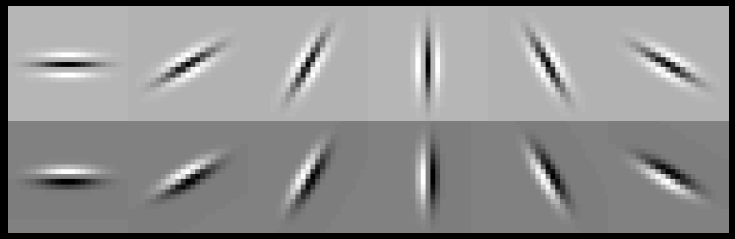


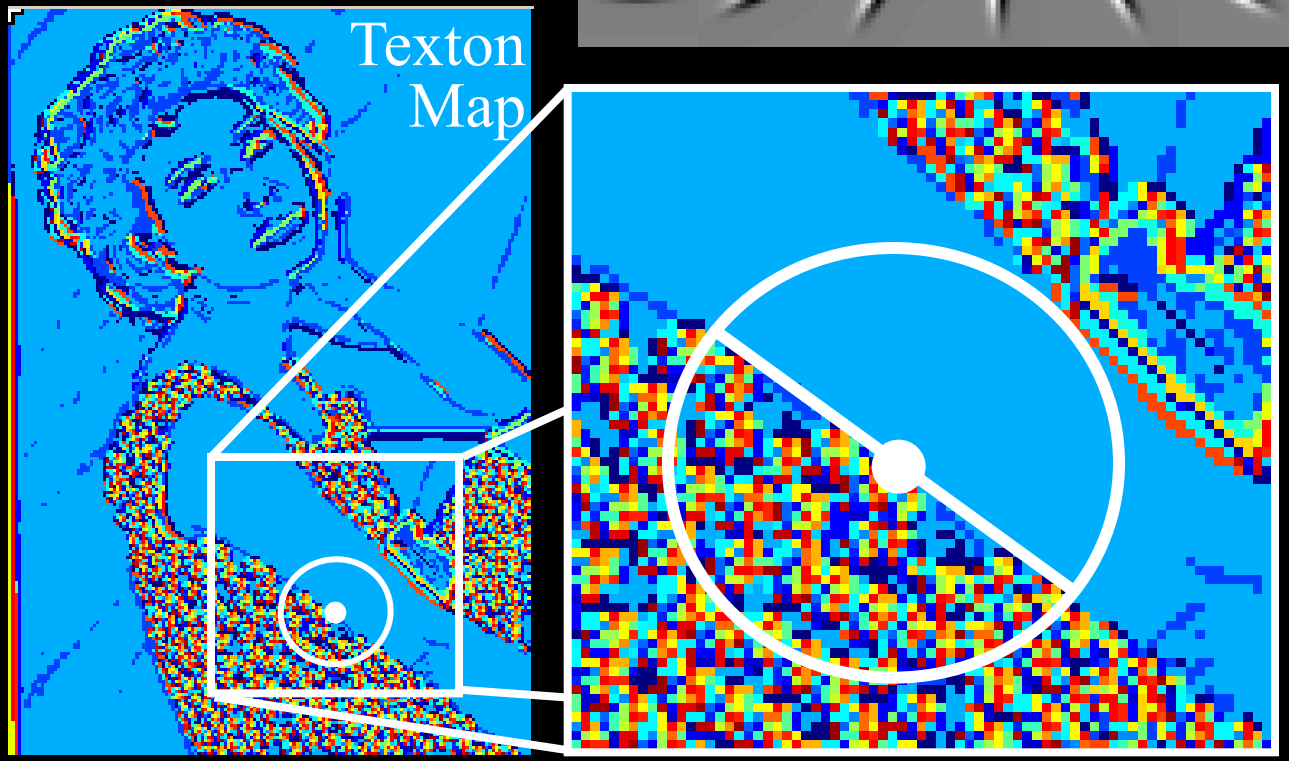
  2) Vector quantization (k-means) on filter outputs;

  3) Quantization centers are the textons.

- Spatial distribution of textons defines the texture;

Texture Feature





- Texture Gradient TG (x,y,r,θ)
  - χ^2 difference of texton histograms
  - Textons are vector-quantized filter outputs

Cue Combination Models

- Classification Trees
  - Top-down splits to maximize entropy, error bounded
- Density Estimation
  - Adaptive bins using k-means
- Logistic Regression, 3 variants
  - Linear and quadratic terms
  - Confidence-rated generalization of AdaBoost (Schapire&Singer)
- Hierarchical Mixtures of Experts (Jordan&Jacobs)
  - Up to 8 experts, initialized top-down, fit with EM
- Support Vector Machines (`libsvm`, Chang&Lin)
  - Gaussian kernel, ν-parameterization

➢ Range over bias, complexity, parametric/non-parametric

Dataflow

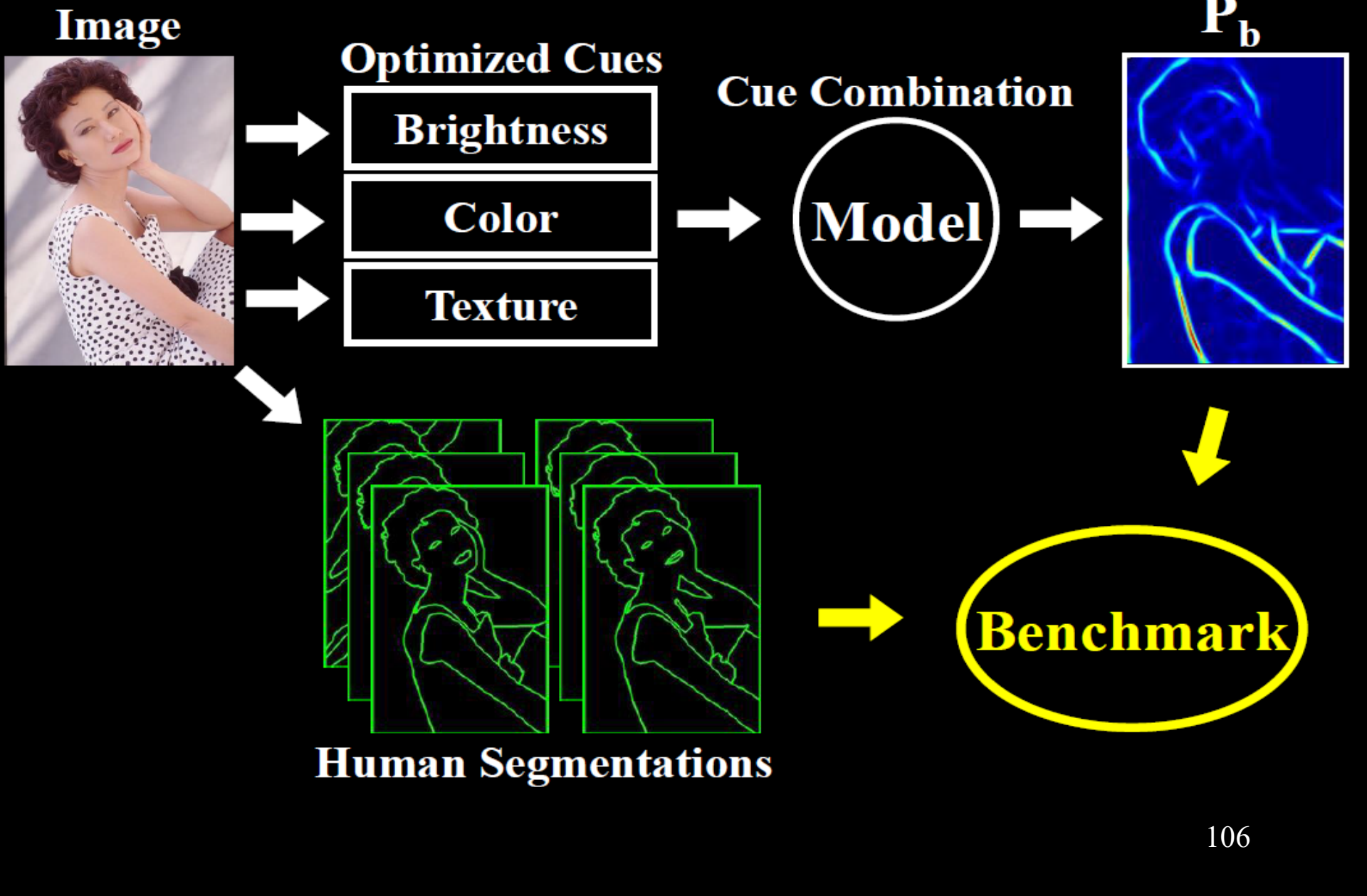


Image
Optimized Cues
Brightness
Color
Texture
Cue Combination
Model
P_b
Human Segmentations
Benchmark

Computing Precision/Recall

Recall = Pr(signal|truth) = fraction of ground truth found by the signal

Precision = Pr(truth|signal) = fraction of signal that is correct

- Always a trade-off between the two
- Standard measures in information retrieval (van Rijsbergen XX)
- ROC from standard signal detection: the wrong approach

Strategy

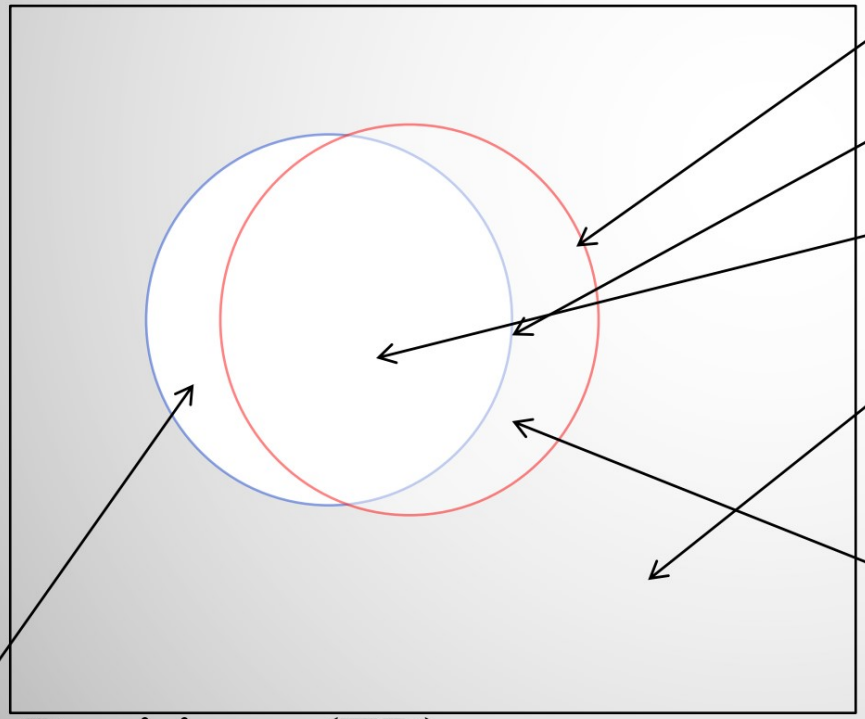
- Detector output (Pb) is a soft boundary map
- Compute precision/recall curve:
  - Threshold Pb at many points t in [0,1]
  - Recall = Pr(Pb>t|seg=1)
  - Precision = Pr(seg=1|Pb>t)

Evaluate Edge Detection

$$\text{precision} = \frac{GT \bigcap RM}{RM} = \frac{TP}{RM}$$

$$\text{recall} = \frac{GT \bigcap RM}{GT} = \frac{TP}{GT}$$

$$F_1 = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$$



Ground Truth (GT)

Results of Method (RM)

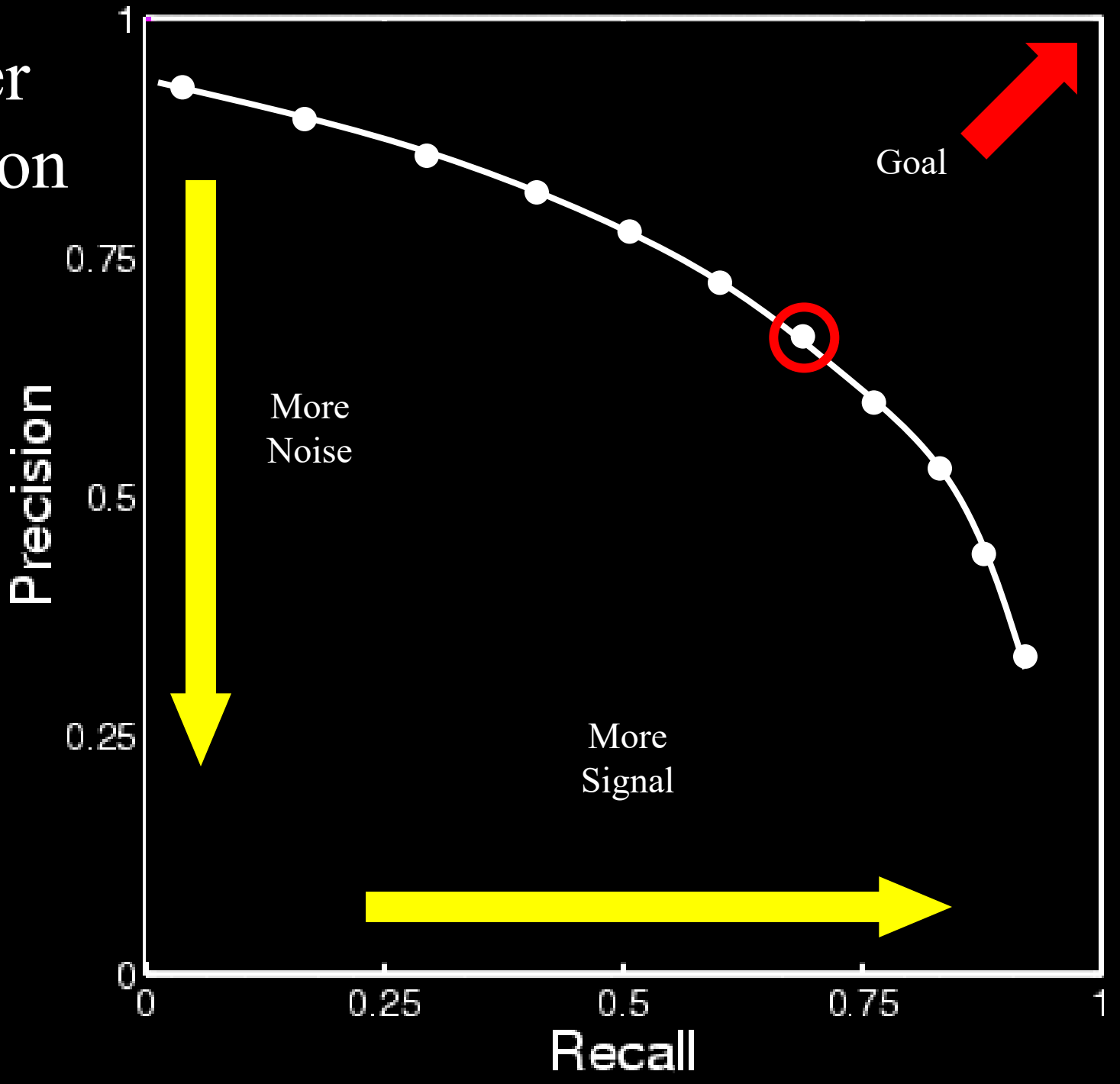
True Positives (TP)

True Negatives (TN)

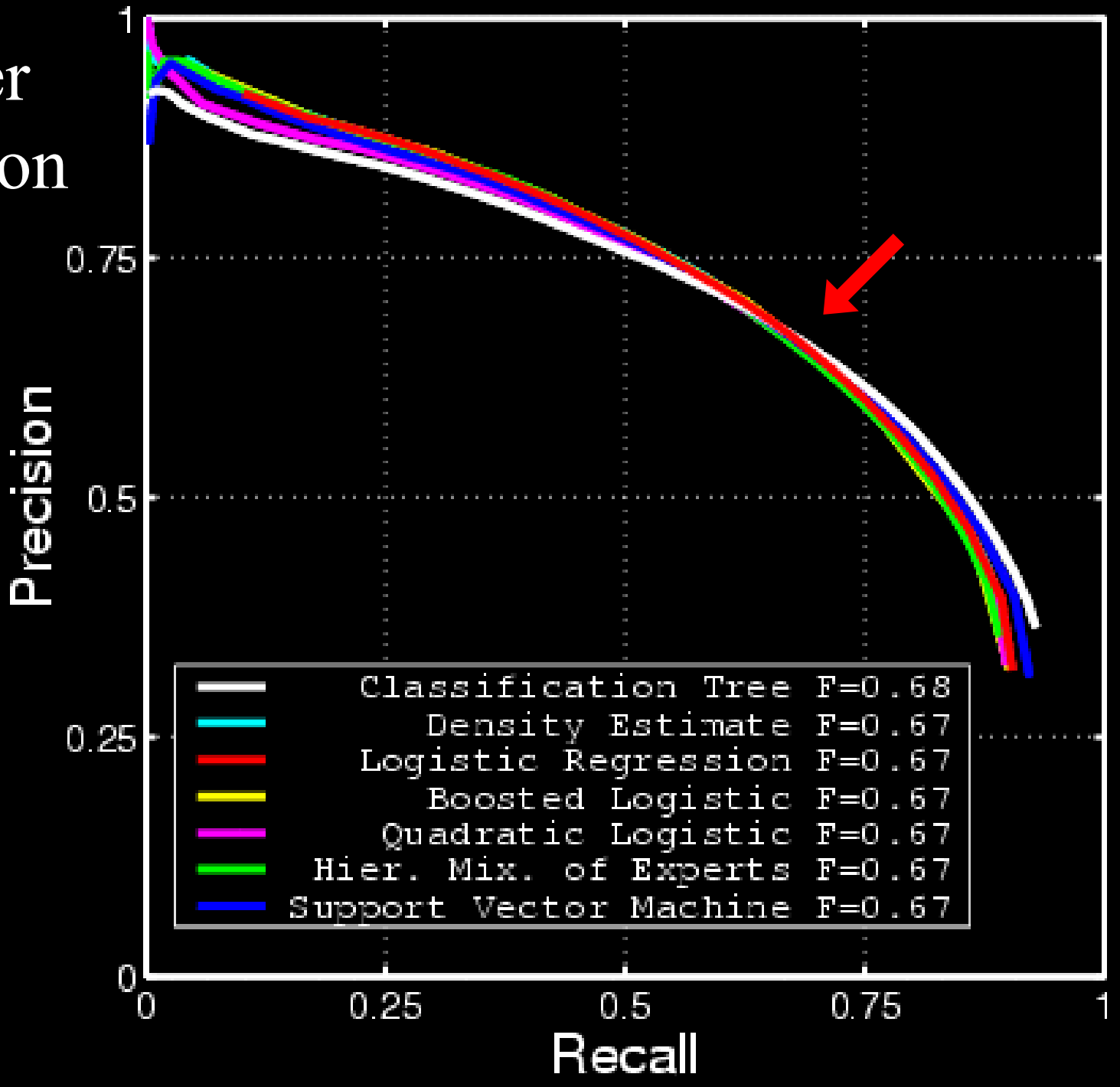
False Negatives (FN)

False Positives (FP)

Classifier Comparison

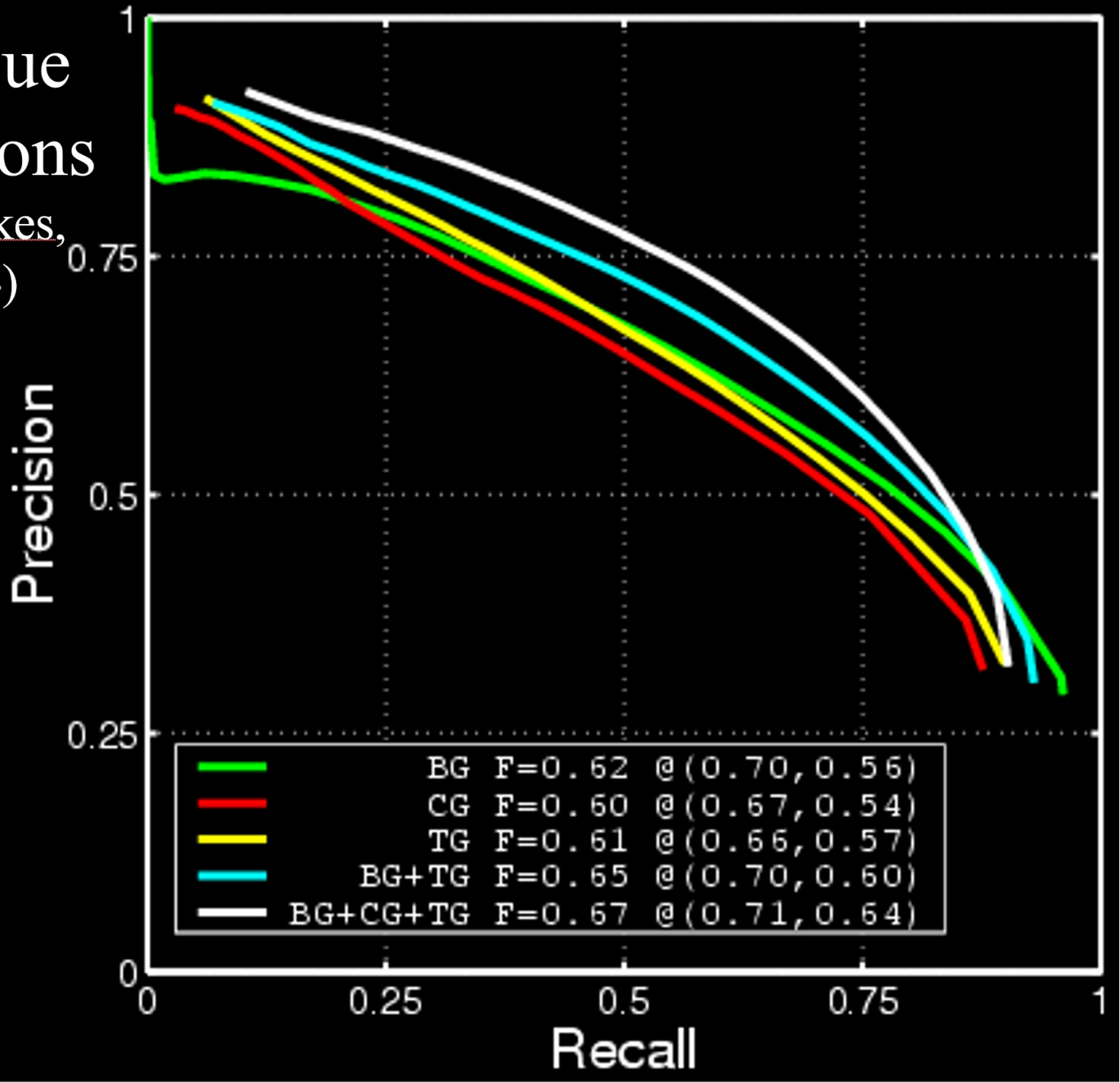


Classifier Comparison



Various Cue Combinations

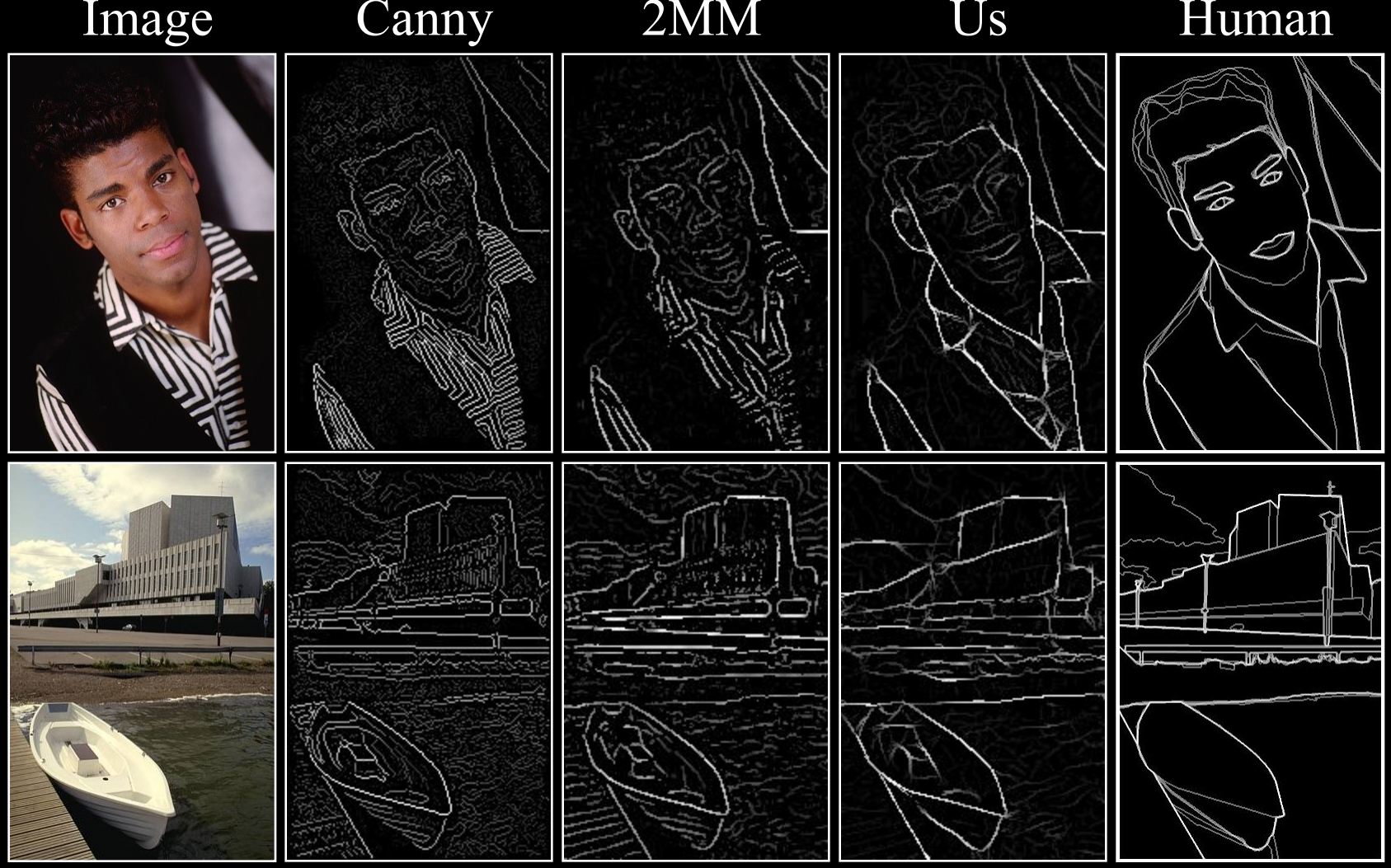
(Martin, Fowlkes, Malik, 2004)



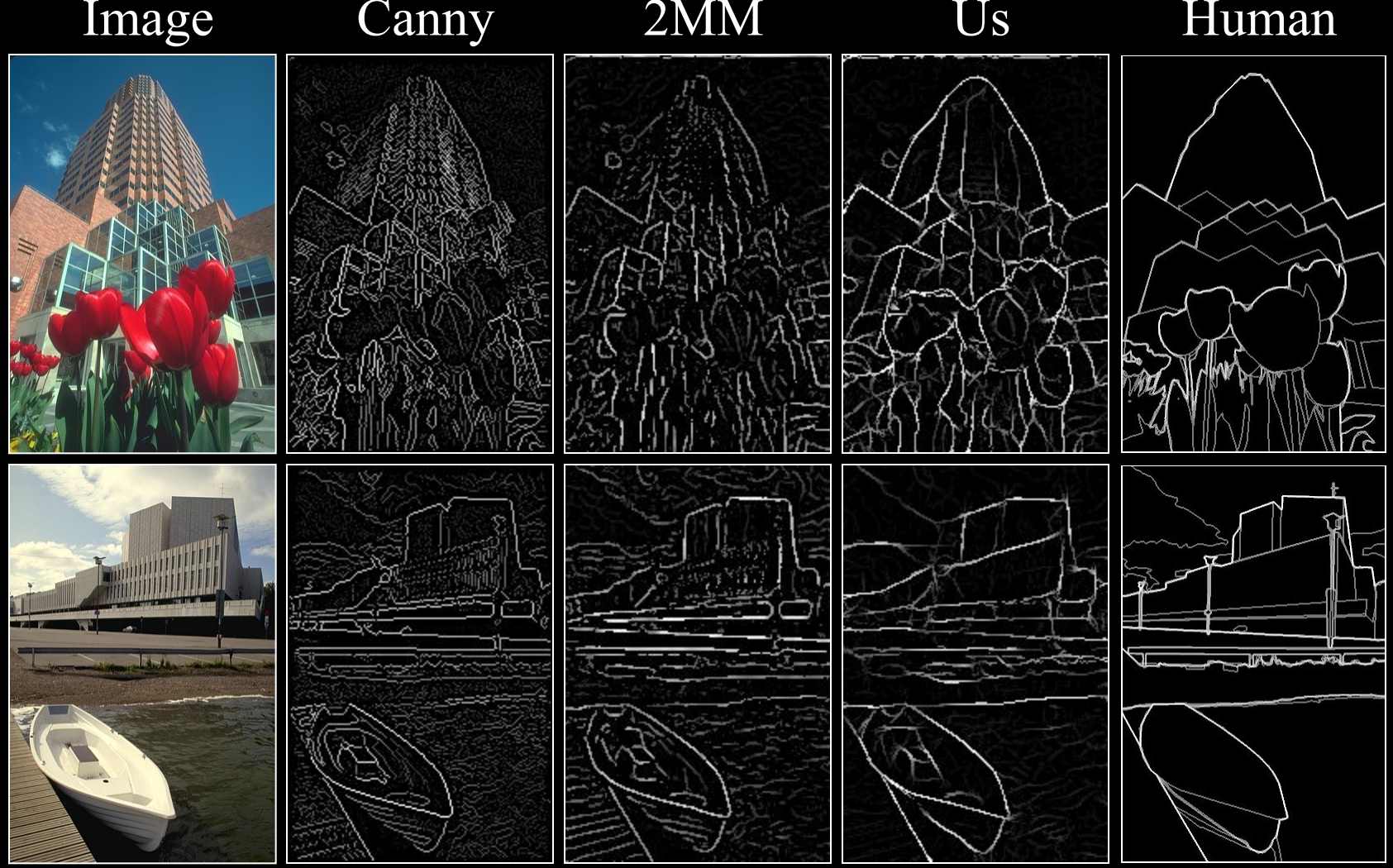
Alternate Approaches

- Canny Detector
  - Canny 1986
  - MATLAB implementation
  - With and without hysteresis
- Second Moment Matrix
  - Nitzberg/Mumford/Shiota 1993
  - cf. Förstner and Harris corner detectors
  - Used by Konishi et al. 1999 in learning framework
  - Logistic model trained on full eigenspectrum

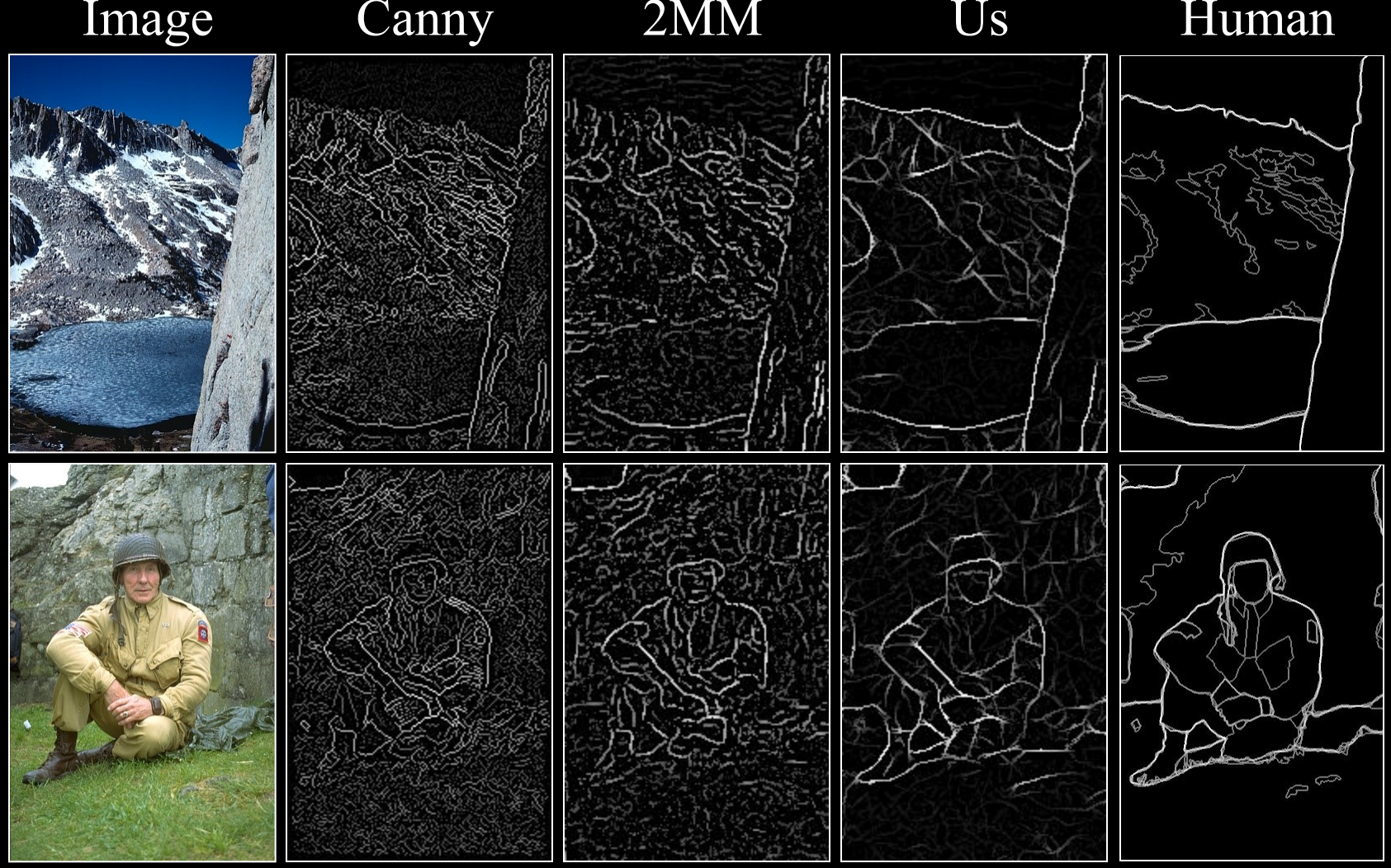
P_b Images



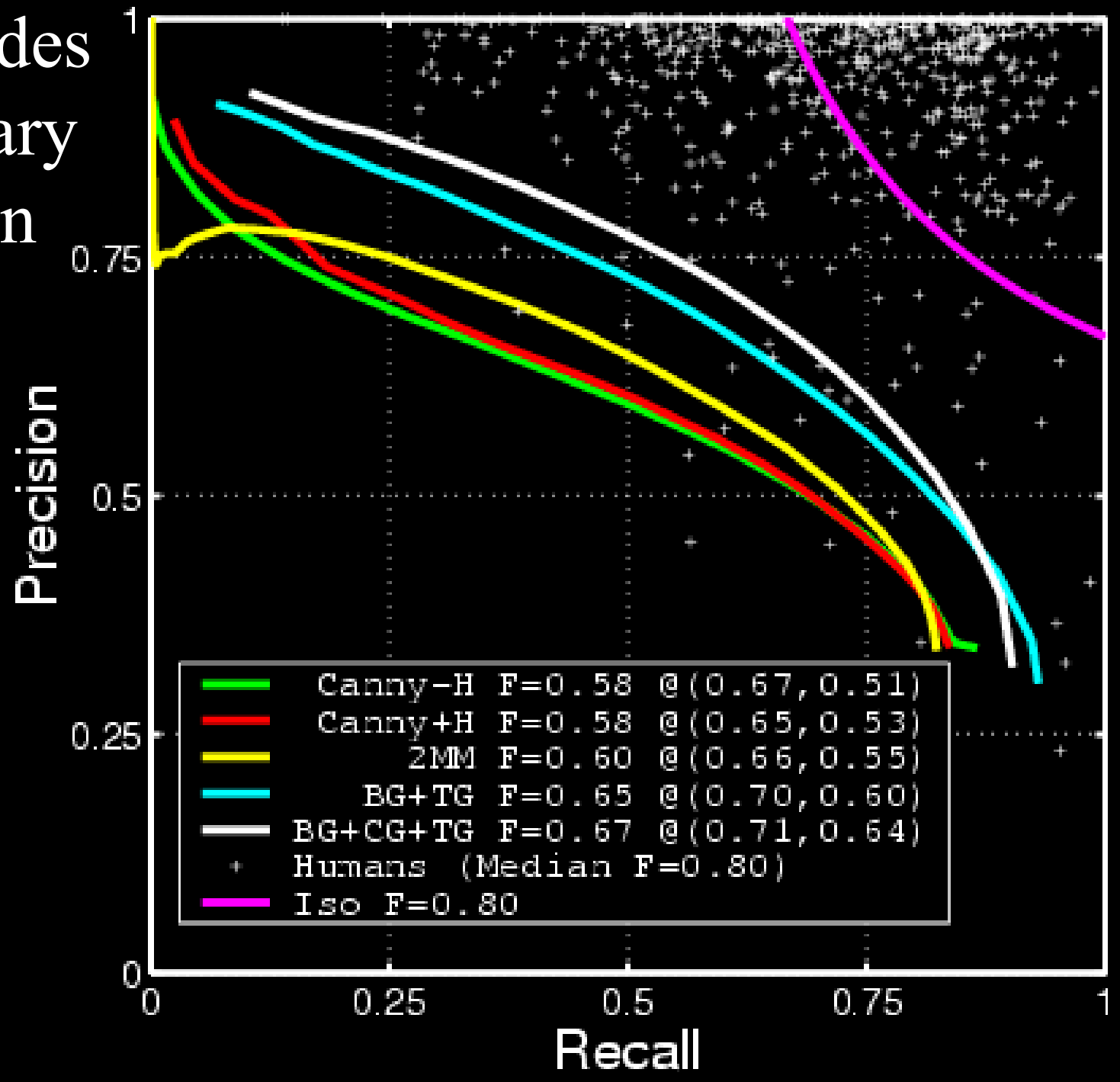
P_b Images II



P_b Images III



Two Decades of Boundary Detection



Findings

1. A simple linear model is sufficient for cue combination
   - All cues weighted approximately equally in logistic
2. Proper texture edge model is not optional for complex natural images
   - Texture suppression is not sufficient!
3. Significant improvement over state-of-the-art in boundary detection
   - $P_b(x,y,\theta)$ useful for higher-level processing
4. Empirical approach critical for both cue calibration and cue combination

Brightness

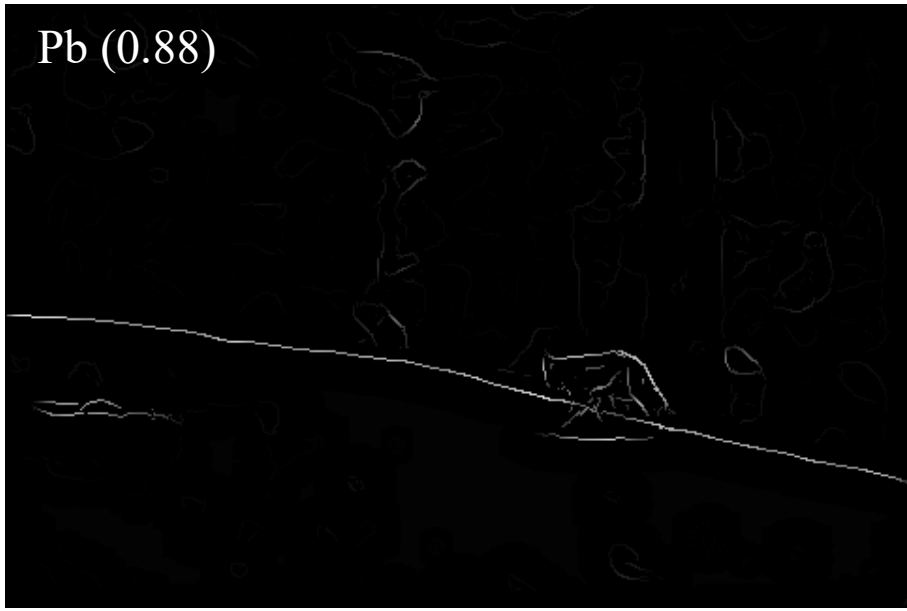
Color

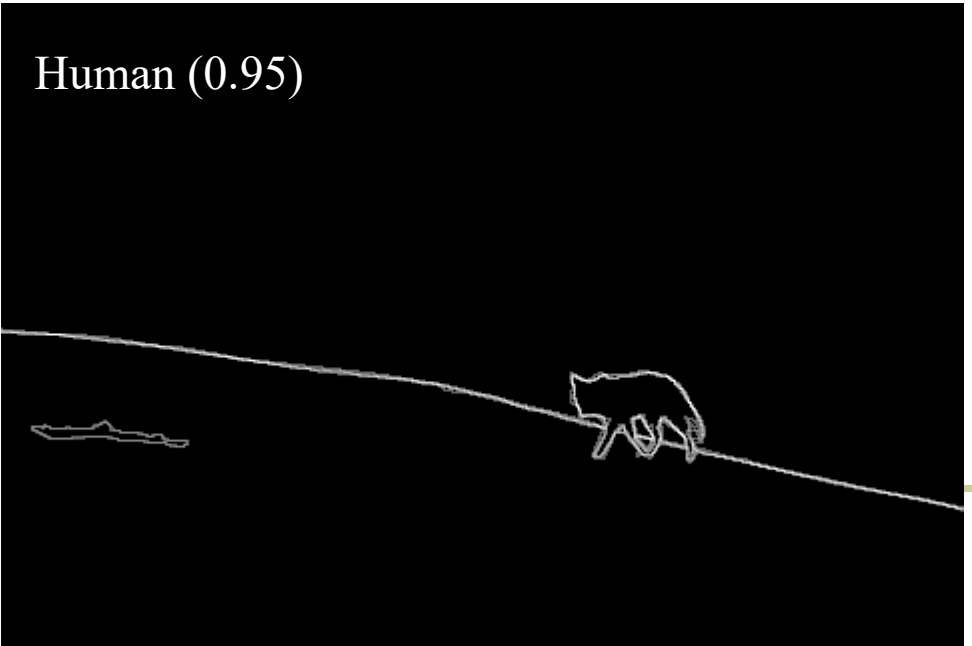
Texture

Combined

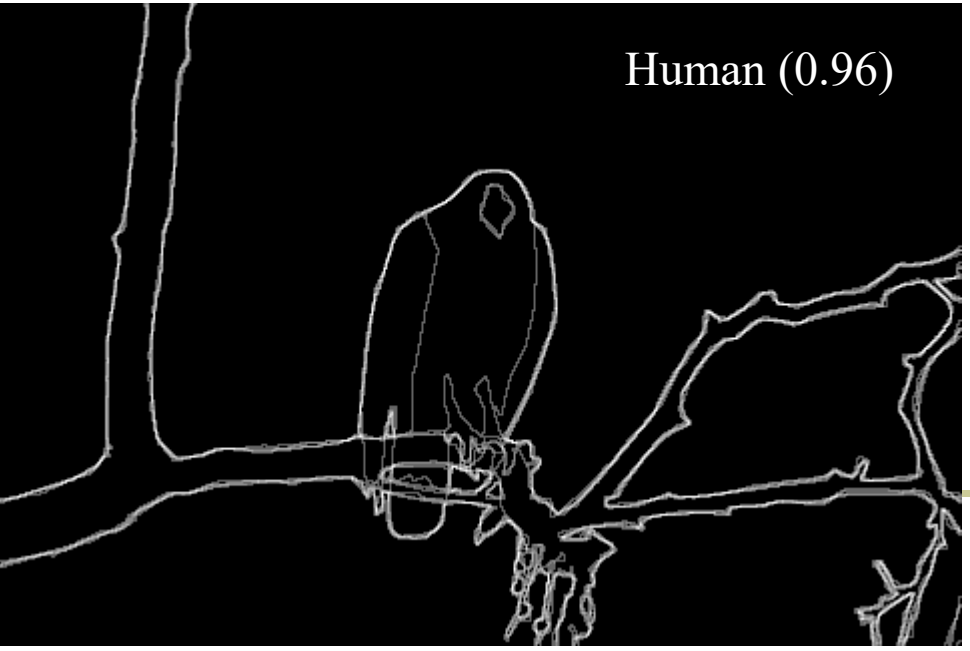
Human

Results

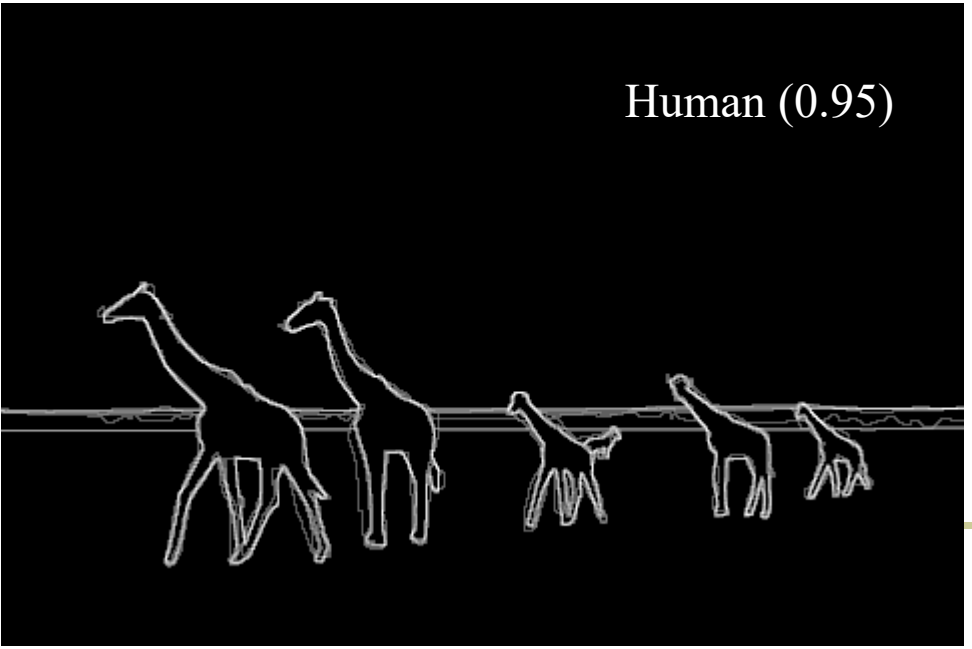


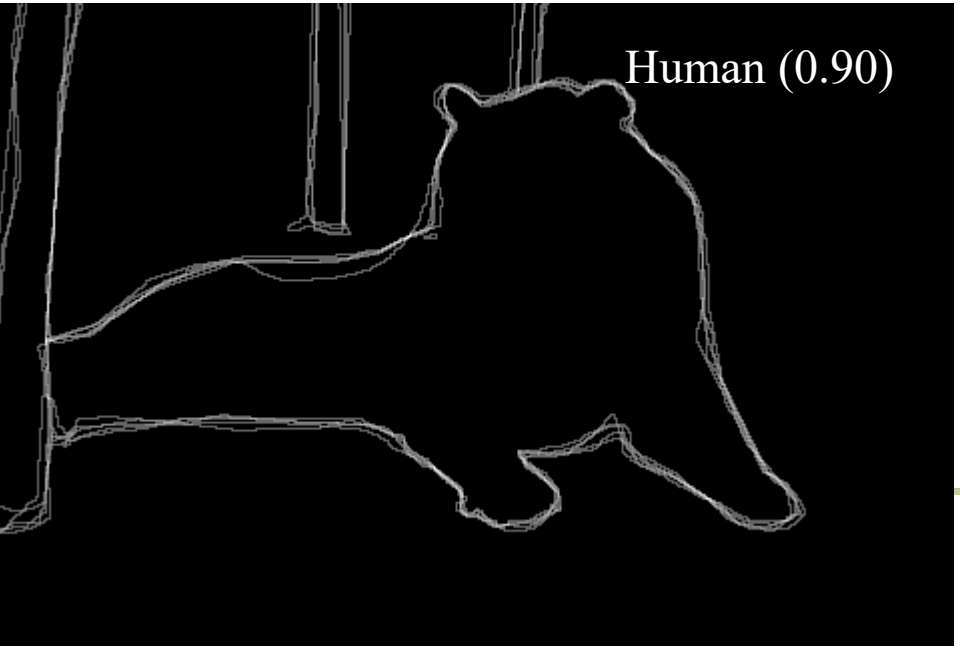


Results




Pb (0.63)


Human (0.95)



For more:
http://www.eecs.berkeley.edu/Research/Projects/CS/vision/bsds/bench/html/108082-color.html

Today's class

- Introduction to edge detection
- Gradients and edges
- Canny edge detector
- Object contour
- Pb edge detector
- **Straight line detection**

Finding straight lines

Finding line segments using connected components

1. Compute canny edges
   - Compute: gx, gy (DoG in x,y directions)
   - Compute: theta = atan(gy / gx)
2. Assign each edge to one of 8 directions
3. For each direction d, get edgelets:
   - find connected components for edge pixels with directions in {d-1, d, d+1}
4. Compute straightness and theta of edgelets using eig of x,y 2nd moment matrix of their points

$$\mathbf{M} = \begin{bmatrix} \sum (x-\mu_x)^2 & \sum (x-\mu_x)(y-\mu_y) \\ \sum (x-\mu_x)(y-\mu_y) & \sum (y-\mu_y)^2 \end{bmatrix} \qquad [v,\lambda] = \text{eig}(\mathbf{M})$$

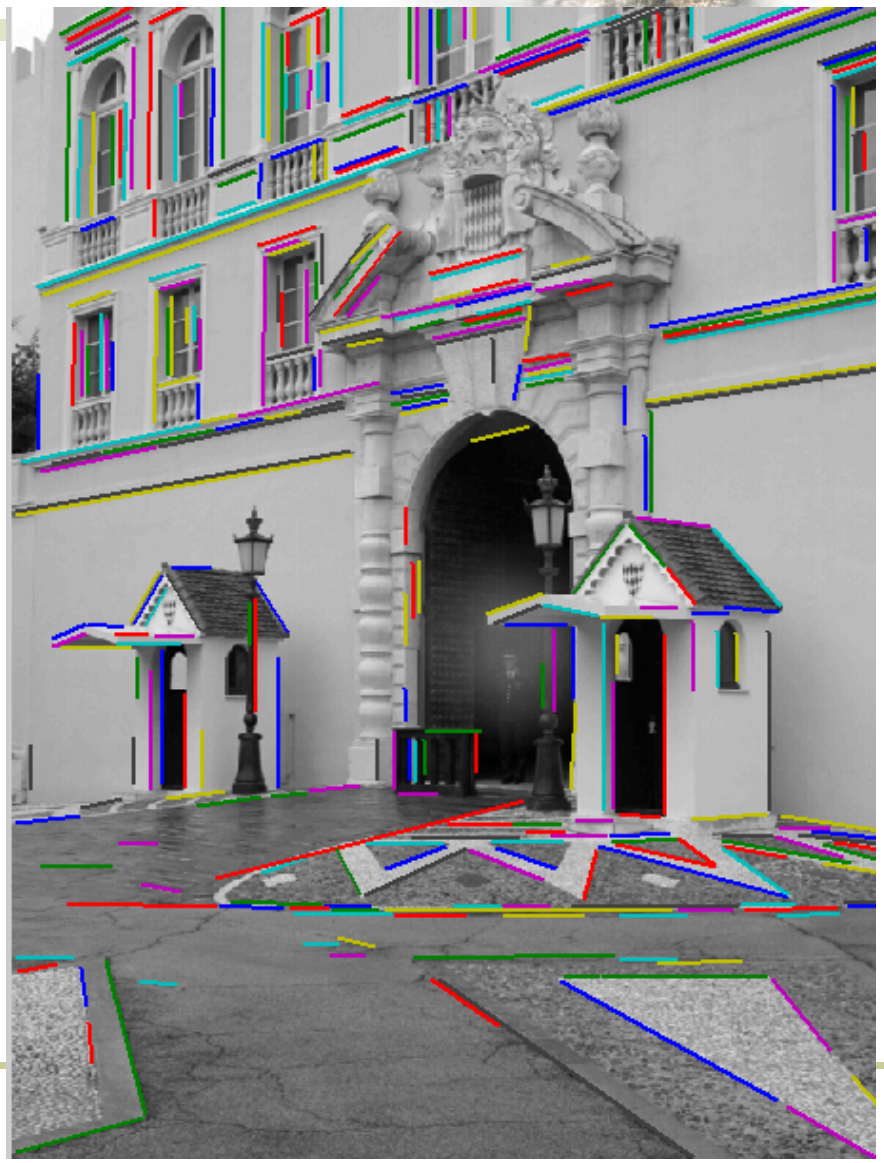
Larger eigenvector

$$\theta = \text{atan}\,2(v(2,2), v(1,2))$$

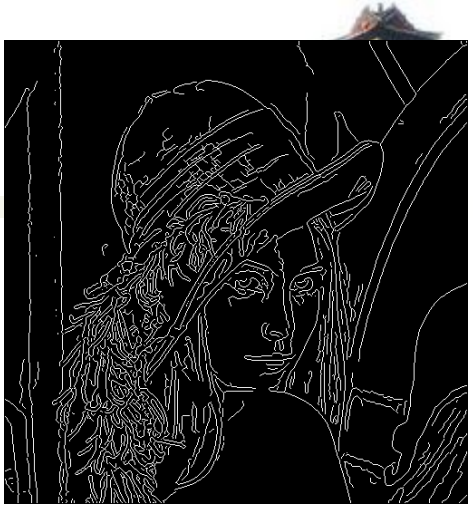
$$conf = \lambda_2 / \lambda_1$$

5. Threshold on straightness, store segment

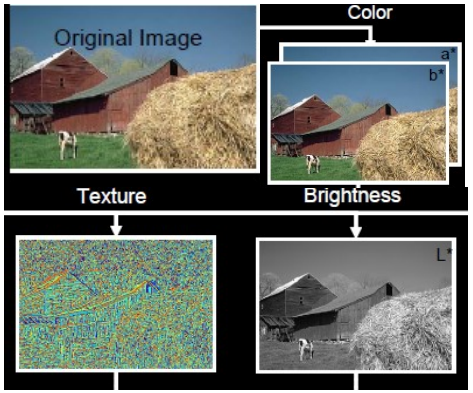
Canny lines → … →straight edges



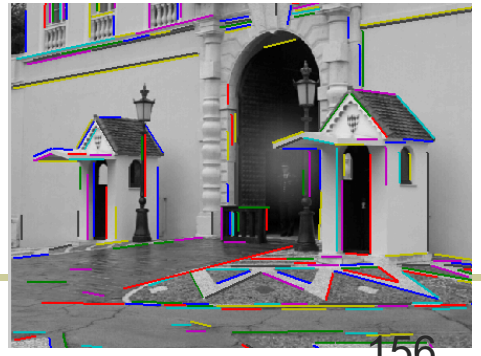
Things to remember



- Canny edge detector = smooth → derivative → thin → threshold → link



- Pb: learns weighting of gradient, color, texture differences
  - More recent learning approaches give at least as good accuracy and are faster



- Straight line detector = canny + gradient orientations → orientation binning → linking → check for straightness